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Methods of detecting bladder cancer

Abstract

Compositions and methods for detecting bladder cancer are provided. In some embodiments, methods of monitoring recurrence of bladder cancer are provided. In some embodiments, the methods comprise detecting a set of markers consisting of CRH, IGF2, KRT20, and ANXA10.

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Background/Summary

1. CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. application Ser. No. 16/272,978, filed Feb. 11, 2019, which is a divisional of U.S. application Ser. No. 14/394,352, filed Oct. 14, 2014, which is a 371 application of International Application No. PCT/US2013/037334, filed Apr. 19, 2013, which claims the benefit of U.S. Provisional Application No. 61/636,194, filed Apr. 20, 2012 and of U.S. Provisional Application No. 61/770,803 filed Feb. 28, 2013, all of which are incorporated herein by reference in their entireties.

2. SEQUENCE LISTING

(1) The present application is filed with a Sequence Listing in electronic format. The Sequence Listing is provided as a file entitled "01148-0006-02US_Sequence_Listing_ST26" created on Sep.

16, 2023, which is 69,632 bytes in size. The information in the electronic format of the sequence listing is incorporated herein by reference in its entirety.

3. FIELD OF THE INVENTION

(2) Compositions and methods for detecting bladder cancer are provided. In particular, bladder cancer markers and panels of markers useful in the detection of bladder cancer are provided.

4. BACKGROUND

(3) 386,000 cases of bladder cancer are diagnosed globally each year, including 70,500 cases per year in the United States. The incidence of bladder cancer is three times higher in men than in women. The highest incidence and prevalence are found in the European Union, North America, North Africa, and the Middle East

(4) Smoking is the greatest risk factor for bladder cancer. Additional risk factors include chemical exposure, chemotherapy (such as Cytosan), radiation treatment, and chronic bladder infection.

(5) Bladder tumors include papillary tumors, which are urothelial carcinomas that grow narrow, finger-like projections; and nonpapillary (sessile) tumors, such as carcinoma-in-situ, which are less common but have a high risk of becoming invasive.

(6) Symptoms of bladder cancer can include abdominal pain, blood in the urine, bone pain or tenderness, fatigue, painful urination, frequent urination, urinary urgency, incontinence, and weight loss. Diagnosis is generally based on imaging, urinalysis, and/or biopsy.

(7) The prognosis for bladder cancer depends on the stage of cancer at diagnosis. The prognosis for early tumors is favorable, while the prognosis for advanced tumors is poor. Long-term follow up is recommended to detect cancer recurrence, which occurs in up to 70% of bladder cancers. For the first two years, cystoscopy and urine cytology are recommended every 3 to 4 months, and then at longer intervals in subsequent years, often for the patient's lifetime. These methods are invasive and costly, making bladder cancer one of the most expensive cancers to treat from diagnosis until death.

(8) Existing non-invasive diagnostic tests include ImmunoCyt™ (Scimedex, Denville, NJ) and UroVysion® (Abbott Molecular, Abbott Park, IL). ImmunoCyt™ is a cytology assay that uses a cocktail of three monoclonal antibodies labeled with fluorescent markers to detect certain cellular markers of bladder cancer in exfoliated cells isolated from urine samples. ImmunoCyt™ is used in conjunction with standard urine cytology to improve cytology's sensitivity at detecting tumor cells. UroVysion® is also a cytology-based assay, which detects aneuploidy in certain chromosomes via fluorescent in situ hybridization (FISH). Determination of the results is conducted by enumerating signals through microscopic examination of the nucleus of cells in urine.

(9) Improved methods for early detection of bladder cancer are needed. In particular, an accurate urine-based diagnostic test that does not rely on cytology could reduce the need for costly and invasive cystoscopy and labor-intensive and potentially subjective cytology assays.

5. SUMMARY

(10) Compositions and methods for detecting bladder cancer are provided. In particular, bladder cancer markers and panels of markers useful in the detection of bladder cancer are provided. In some embodiments, the levels of CRH, IGF2, KRT20 and ANXA10 mRNA are measured, for example, by quantitative RT-PCR, and the results can be used to determine whether or not a subject has bladder cancer. In some embodiments, the levels of CRH, IGF2, KRT20, and ANXA10 mRNA are normalized to an endogenous control. In some embodiments, the endogenous control is ABL mRNA. In some embodiments, an endogenous control is selected that is expected to be expressed at similar levels in bladder urothelial cells from subjects with and without bladder cancer. In some embodiments, the sample is a urine sample. In some embodiments, the present methods are used to monitor subjects with a history of bladder cancer for tumor recurrence. In some embodiments, the subject has been treated with Bacillus Calmette-Guerin (BCG) within the past three months. In some embodiments, the present methods are used to detect bladder cancer in subjects with no history of bladder cancer. In some such embodiments, the subjects have symptoms of bladder cancer. Nonlimiting exemplary symptoms of bladder cancer include abdominal pain, blood in the

urine, bone pain or tenderness, fatigue, painful urination, frequent urination, urinary urgency, incontinence, and weight loss.

(11) In some embodiments, methods for detecting the presence of bladder cancer in a subject are provided. In some embodiments, a method comprises detecting the levels of each marker of a set of bladder cancer markers in a sample from the subject, wherein the set of bladder cancer markers consists of corticotrophin releasing hormone (CRH), insulin-like growth factor 2 (IGF2), keratin 20 (KRT20) and annexin 10 (ANXA10). In some embodiments, detection of an elevated level of at least one marker indicates the presence of bladder cancer in the subject. In some embodiments, a method further comprises detecting an endogenous control. In some embodiments, the endogenous control is selected from ABL, GUSB, GAPDH, TUBB, and UPK1. In some embodiments, the endogenous control is ABL. In some embodiments, a method comprises detecting an exogenous control. In some embodiments, the exogenous control is an RNA. In some such embodiments, the exogenous control is an Armored RNA®.

(12) In some embodiments, detecting comprises RT-PCR. In some embodiments, detecting comprises quantitative RT-PCR. In some embodiments, a method comprises comparing a Ct value or a Δ Ct value to a threshold Ct value or Δ Ct value. In some embodiments, Δ Ct is the Ct value for the endogenous control minus the Ct value for the marker. In some embodiments, the RT-PCR reaction takes less than three hours or less than 2 hours from an initial denaturation step through a final extension step.

(13) In some embodiments, a method comprises contacting RNA from the sample with a set of bladder cancer marker primer pairs, wherein the set of bladder cancer marker primer pairs consists of a first primer pair for detecting CRH, a second primer pair for detecting IGF2, a third primer pair for detecting KRT20, and a fourth primer pair for detecting ANXA10. In some embodiments, the first primer pair comprises a first primer comprising SEQ ID NO: 19 and a second primer comprising SEQ ID NO: 20, wherein each primer is less than 50, less than 45, less than 40, less than 35, or less than 30 nucleotides long. In some embodiments, the first primer pair comprises a first primer comprising SEQ ID NO: 35 and a second primer comprising SEQ ID NO: 36, wherein each primer is less than 50, less than 45, less than 40, less than 35, or less than 30 nucleotides long. In some embodiments, the second primer pair comprises a first primer comprising SEQ ID NO: 16 and a second primer comprising SEQ ID NO: 17, wherein each primer is less than 50, less than 45, less than 40, less than 35, or less than 30 nucleotides long. In some embodiments, the second primer pair comprises a first primer comprising SEQ ID NO: 32 and a second primer comprising SEQ ID NO: 33, wherein each primer is less than 50, less than 45, less than 40, less than 35, or less than 30 nucleotides long. In some embodiments, the third primer pair comprises a first primer comprising SEQ ID NO: 13 and a second primer comprising SEQ ID NO: 14, wherein each primer is less than 50, less than 45, less than 40, less than 35, or less than 30 nucleotides long. In some embodiments, the third primer pair comprises a first primer comprising SEQ ID NO: 29 and a second primer comprising SEQ ID NO: 30, wherein each primer is less than 50, less than 45, less than 40, less than 35, or less than 30 nucleotides long. In some embodiments, the fourth primer pair comprises a first primer comprising SEQ ID NO: 26 and a second primer comprising SEQ ID NO: 27, wherein each primer is less than 50, less than 45, less than 40, less than 35, or less than 30 nucleotides long. In some embodiments, the fourth primer pair comprises a first primer comprising SEQ ID NO: 38 and a second primer comprising SEQ ID NO: 39, wherein each primer is less than 50, less than 45, less than 40, less than 35, or less than 30 nucleotides long. In some embodiments, the fourth primer pair comprises a first primer comprising SEQ ID NO: 48 and a second primer comprising SEQ ID NO: 39, wherein each primer is less than 50, less than 45, less than 40, less than 35, or less than 30 nucleotides long.

(14) In some embodiments, a method comprises contacting RNA from the sample with a set of bladder cancer marker primer pairs, wherein the set of bladder cancer marker primer pairs consists of a first primer pair for detecting CRH, a second primer pair for detecting IGF2, a third primer pair

[illegible]

nucleotides long.

(15) In some embodiments, a method further comprises contacting RNA from the sample with an endogenous control primer pair. In some such embodiments, the endogenous control primer pair is for detecting an endogenous control is selected from ABL, GUSB, GAPDH, TUBB, and UPK1. In some embodiments, the endogenous control primer pair is for detecting ABL. In some embodiments, the endogenous control primer pair comprises a first primer comprising SEQ ID NO: 8 and a second primer comprising SEQ ID NO: 9, wherein each primer is less than 50, less than 45, less than 40, less than 35, or less than 30 nucleotides long. In some embodiments, the endogenous control primer pair comprises a first primer comprising SEQ ID NO: 41 and a second primer comprising SEQ ID NO: 42, wherein each primer is less than 50, less than 45, less than 40, less than 35, or less than 30 nucleotides long. In some embodiments, the endogenous control primer pair comprises a first primer comprising at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, or at least 15, at least 16, at least 17, or at least 18 nucleotides of SEQ ID NO: 8 and a second primer comprising at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, or at least 15, at least 16, at least 17, or at least 18 nucleotides of SEQ ID NO: 9, wherein each primer is less than 50, less than 45, less than 40, less than 35, or less than 30 nucleotides long. In some embodiments, the endogenous control primer pair comprises a first primer comprising at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, or at least 15, at least 16, at least 17, or at least 18 nucleotides of SEQ ID NO: 41 and a second primer comprising at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, or at least 15, at least 16, at least 17, or at least 18 nucleotides of SEQ ID NO: 42, wherein each primer is less than 50, less than 45, less than 40, less than 35, or less than 30 nucleotides long.

(16) In some embodiments, a method comprises contacting RNA from the sample with an exogenous control primer pair. In some such embodiments, the exogenous control primer pair is for detecting an exogenous RNA. In some embodiments, the exogenous control primer pair comprises a first primer comprising SEQ ID NO: 23 and a second primer comprising SEQ ID NO: 24, wherein each primer is less than 50, less than 45, less than 40, less than 35, or less than 30 nucleotides long. In some embodiments, the exogenous control primer pair comprises a first primer comprising SEQ ID NO: 44 and a second primer comprising SEQ ID NO: 45, wherein each primer is less than 50, less than 45, less than 40, less than 35, or less than 30 nucleotides long. In some embodiments, the exogenous control primer pair comprises a first primer comprising at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, or at least 15, at least 16, at least 17, or at least 18 nucleotides of SEQ ID NO: 23 and a second primer comprising at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, or at least 15, at least 16, at least 17, or at least 18 nucleotides of SEQ ID NO: 24, wherein each primer is less than 50, less than 45, less than 40, less than 35, or less than 30 nucleotides long. In some embodiments, the exogenous control primer pair comprises a first primer comprising at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, or at least 15, at least 16, at least 17, or at least 18 nucleotides of SEQ ID NO: 44 and a second primer comprising at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, or at least 15, at least 16, at least 17, or at least 18 nucleotides of SEQ ID NO: 45, wherein each primer is less than 50, less than 45, less than 40, less than 35, or less than 30 nucleotides long.

(17) In some embodiments, the method comprises forming a set of bladder cancer marker amplicons, wherein the set of bladder cancer marker amplicons consists of a CRH amplicon, an IGF2 amplicon, a KRT20 amplicon, and an ANXA10 amplicon, and contacting the bladder cancer marker amplicons with a set of bladder cancer marker probes, wherein the set of bladder cancer marker probes consists of a first probe for detecting the CRH amplicon, a second probe for detecting the IGF2 amplicon, a third probe for detecting the KRT20 amplicon, and a fourth probe for detecting the ANXA10 amplicon. In some embodiments, the first probe comprises SEQ ID NO: 21 or SEQ ID NO: 37, wherein the first probe is less than 50, less than 45, less than 40, less than

35, or less than 30 nucleotides long. In some embodiments, the second probe comprises SEQ ID NO: 34 or SEQ ID NO: 18, wherein the second probe is less than 50, less than 45, less than 40, less than 35, or less than 30 nucleotides long. In some embodiments, the third probe comprises SEQ ID NO: 15 or SEQ ID NO: 31, wherein the third probe is less than 50, less than 45, less than 40, less than 35, or less than 30 nucleotides long. In some embodiments, the fourth probe comprises SEQ ID NO: 28 or SEQ ID NO: 40, wherein the fourth probe is less than 50, less than 45, less than 40, less than 35, or less than 30 nucleotides long. In some embodiments, the first probe comprises at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, or at least 15, at least 16, at least 17, or at least 18 nucleotides of SEQ ID NO: 21 or at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, or at least 15, at least 16, at least 17, or at least 18 nucleotides of SEQ ID NO: 37, wherein the first probe is less than 50, less than 45, less than 40, less than 35, or less than 30 nucleotides long. In some embodiments, the second probe comprises at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, or at least 15, at least 16, at least 17, or at least 18 nucleotides of SEQ ID NO: 34 or at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, or at least 15, at least 16, at least 17, or at least 18 nucleotides of SEQ ID NO: 18, wherein the second probe is less than 50, less than 45, less than 40, less than 35, or less than 30 nucleotides long. In some embodiments, the third probe comprises at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, or at least 15, at least 16, at least 17, or at least 18 nucleotides of SEQ ID NO: 15 or at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, or at least 15, at least 16, at least 17, or at least 18 nucleotides of SEQ ID NO: 31, wherein the third probe is less than 50, less than 45, less than 40, less than 35, or less than 30 nucleotides long. In some embodiments, the fourth probe comprises at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, or at least 15, at least 16, at least 17, or at least 18 nucleotides of SEQ ID NO: 28 or at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, or at least 15, at least 16, at least 17, or at least 18 nucleotides of SEQ ID NO: 40, wherein the fourth probe is less than 50, less than 45, less than 40, less than 35, or less than 30 nucleotides long. In some embodiments, each bladder cancer marker probe comprises a dye, and wherein each dye is detectably different from the other three labels. In some embodiments, each bladder cancer marker probe comprises a fluorescent dye and a quencher molecule.

(18) In some embodiments, a method comprises forming an endogenous control amplicon, and contacting the endogenous control amplicon with an endogenous control probe. In some embodiments, the endogenous control probe comprises at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, or at least 15, at least 16, at least 17, or at least 18 nucleotides of SEQ ID NO: 10, 11, 12, or 43, wherein the endogenous control probe is less than 50, less than 45, less than 40, less than 35, or less than 30 nucleotides long. In some embodiments, the endogenous control probe comprises a dye that is detectably different from the dyes of the bladder cancer marker probes.

(19) In some embodiments, a method comprises forming an exogenous control amplicon, and contacting the exogenous control amplicon with an exogenous control probe. In some embodiments, the exogenous control probe comprises at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, or at least 15, at least 16, at least 17, or at least 18 nucleotides of SEQ ID NO: 25 or 46, wherein the exogenous control probe is less than 50, less than 45, less than 40, less than 35, or less than 30 nucleotides long. In some embodiments, the exogenous control probe comprises a dye that is detectably different from the dyes of the bladder cancer marker probes and the endogenous control probe.

(20) In some embodiments, the set of bladder cancer markers are detected in a single multiplex reaction.

(21) In some embodiments, the sample comprises urothelial cells. In some embodiments, the sample is selected from a urine sample and a bladder washing sample. In some embodiments, the subject has a history of bladder cancer. In some embodiments, the subject is being monitored for

recurrence of bladder cancer.

(22) In some embodiments, compositions are provided. In some embodiments, a composition comprises a set of bladder cancer marker primer pairs, wherein the set of bladder cancer marker primer pairs consists of a first primer pair for detecting CRH, a second primer pair for detecting IGF2, a third primer pair for detecting KRT20, and a fourth primer pair for detecting ANXA10. In some embodiments, the first primer pair comprises a first primer comprising SEQ ID NO: 19 and a second primer comprising SEQ ID NO: 20, wherein each primer is less than 50, less than 45, less than 40, less than 35, or less than 30 nucleotides long. In some embodiments, the first primer pair comprises a first primer comprising SEQ ID NO: 35 and a second primer comprising SEQ ID NO: 36, wherein each primer is less than 50, less than 45, less than 40, less than 35, or less than 30 nucleotides long. In some embodiments, the second primer pair comprises a first primer comprising SEQ ID NO: 16 and a second primer comprising SEQ ID NO: 17, wherein each primer is less than 50, less than 45, less than 40, less than 35, or less than 30 nucleotides long. In some embodiments, the second primer pair comprises a first primer comprising SEQ ID NO: 32 and a second primer comprising SEQ ID NO: 33, wherein each primer is less than 50, less than 45, less than 40, less than 35, or less than 30 nucleotides long. In some embodiments, the third primer pair comprises a first primer comprising SEQ ID NO: 13 and a second primer comprising SEQ ID NO: 14, wherein each primer is less than 50, less than 45, less than 40, less than 35, or less than 30 nucleotides long. In some embodiments, the third primer pair comprises a first primer comprising SEQ ID NO: 29 and a second primer comprising SEQ ID NO: 30, wherein each primer is less than 50, less than 45, less than 40, less than 35, or less than 30 nucleotides long. In some embodiments, the fourth primer pair comprises a first primer comprising SEQ ID NO: 26 and a second primer comprising SEQ ID NO: 27, wherein each primer is less than 50, less than 45, less than 40, less than 35, or less than 30 nucleotides long. In some embodiments, the fourth primer pair comprises a first primer comprising SEQ ID NO: 38 and a second primer comprising SEQ ID NO: 39, wherein each primer is less than 50, less than 45, less than 40, less than 35, or less than 30 nucleotides long. In some embodiments, the fourth primer pair comprises a first primer comprising SEQ ID NO: 48 and a second primer comprising SEQ ID NO: 39, wherein each primer is less than 50, less than 45, less than 40, less than 35, or less than 30 nucleotides long.

(23) In some embodiments, a composition comprises a set of bladder cancer marker primer pairs, wherein the set of bladder cancer marker primer pairs consists of a first primer pair for detecting CRH, a second primer pair for detecting IGF2, a third primer pair for detecting KRT20, and a fourth primer pair for detecting ANXA10. In some embodiments, the first primer pair comprises a first primer comprising at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, or at least 15, at least 16, at least 17, or at least 18 nucleotides of SEQ ID NO: 19 and a second primer comprising at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, or at least 15, at least 16, at least 17, or at least 18 nucleotides of SEQ ID NO: 20, wherein each primer is less than 50, less than 45, less than 40, less than 35, or less than 30 nucleotides long. In some embodiments, the first primer pair comprises a first primer comprising at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, or at least 15, at least 16, at least 17, or at least 18 nucleotides of SEQ ID NO: 35 and a second primer comprising at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, or at least 15, at least 16, at least 17, or at least 18 nucleotides of SEQ ID NO: 36, wherein each primer is less than 50, less than 45, less than 40, less than 35, or less than 30 nucleotides long. In some embodiments, the second primer pair comprises a first primer comprising at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, or at least 15, at least 16, at least 17, or at least 18 nucleotides of SEQ ID NO: 16 and a second primer comprising at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, or at least 15, at least 16, at least 17, or at least 18 nucleotides of SEQ ID NO: 17, wherein each primer is less than 50, less than 45, less than 40, less than 35, or less than 30 nucleotides long. In some embodiments, the second primer pair comprises a first primer comprising at least 8, at least 9,

at least 10, at least 11, at least 12, at least 13, at least 14, or at least 15, at least 16, at least 17, or at least 18 nucleotides of SEQ ID NO: 32 and a second primer comprising at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, or at least 15, at least 16, at least 17, or at least 18 nucleotides of SEQ ID NO: 33, wherein each primer is less than 50, less than 45, less than 40, less than 35, or less than 30 nucleotides long. In some embodiments, the third primer pair comprises a first primer comprising at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, or at least 15, at least 16, at least 17, or at least 18 nucleotides of SEQ ID NO: 13 and a second primer comprising at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, or at least 15, at least 16, at least 17, or at least 18 nucleotides of SEQ ID NO: 14, wherein each primer is less than 50, less than 45, less than 40, less than 35, or less than 30 nucleotides long. In some embodiments, the third primer pair comprises a first primer comprising at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, or at least 15, at least 16, at least 17, or at least 18 nucleotides of SEQ ID NO: 29 and a second primer comprising at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, or at least 15, at least 16, at least 17, or at least 18 nucleotides of SEQ ID NO: 30, wherein each primer is less than 50, less than 45, less than 40, less than 35, or less than 30 nucleotides long. In some embodiments, the fourth primer pair comprises a first primer comprising at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, or at least 15, at least 16, at least 17, or at least 18 nucleotides of SEQ ID NO: 26 and a second primer comprising at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, or at least 15, at least 16, at least 17, or at least 18 nucleotides of SEQ ID NO: 27, wherein each primer is less than 50, less than 45, less than 40, less than 35, or less than 30 nucleotides long. In some embodiments, the fourth primer pair comprises a first primer comprising at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, or at least 15, at least 16, at least 17, or at least 18 nucleotides of SEQ ID NO: 38 and a second primer comprising at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, or at least 15, at least 16, at least 17, or at least 18 nucleotides of SEQ ID NO: 39, wherein each primer is less than 50, less than 45, less than 40, less than 35, or less than 30 nucleotides long. In some embodiments, the fourth primer pair comprises a first primer comprising at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, or at least 15, at least 16, at least 17, or at least 18 nucleotides of SEQ ID NO: 48 and a second primer comprising at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, or at least 15, at least 16, at least 17, or at least 18 nucleotides of SEQ ID NO: 39, wherein each primer is less than 50, less than 45, less than 40, less than 35, or less than 30 nucleotides long.

(24) In some embodiments, a composition further comprises a set of bladder cancer marker probes, wherein the set of bladder cancer marker probes consists of a first probe for detecting a CRH amplicon, a second probe for detecting an IGF2 amplicon, a third probe for detecting a KRT20 amplicon, and a fourth probe for detecting an ANXA10 amplicon. In some embodiments, the first probe comprises SEQ ID NO: 21 or SEQ ID NO: 37, wherein the first probe is less than 50, less than 45, less than 40, less than 35, or less than 30 nucleotides long. In some embodiments, the second probe comprises SEQ ID NO: 34 or SEQ ID NO: 18, wherein the second probe is less than 50, less than 45, less than 40, less than 35, or less than 30 nucleotides long. In some embodiments, the third probe comprises SEQ ID NO: 15 or SEQ ID NO: 31, wherein the third probe is less than 50, less than 45, less than 40, less than 35, or less than 30 nucleotides long. In some embodiments, the fourth probe comprises SEQ ID NO: 28 or SEQ ID NO: 40, wherein the fourth probe is less than 50, less than 45, less than 40, less than 35, or less than 30 nucleotides long. In some embodiments, the first probe comprises at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, or at least 15, at least 16, at least 17, or at least 18 nucleotides of SEQ ID NO: 21 or at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, or at least 15, at least 16, at least 17, or at least 18 nucleotides of SEQ ID NO: 37, wherein the first probe is less than 50, less than 45, less than 40, less than 35, or less than 30 nucleotides long. In some embodiments, the second probe comprises at least 8, at least 9, at least 10, at least 11, at least 12, at

least 13, at least 14, or at least 15, at least 16, at least 17, or at least 18 nucleotides of SEQ ID NO: 34 or at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, or at least 15, at least 16, at least 17, or at least 18 nucleotides of SEQ ID NO: 18, wherein the second probe is less than 50, less than 45, less than 40, less than 35, or less than 30 nucleotides long. In some embodiments, the third probe comprises at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, or at least 15, at least 16, at least 17, or at least 18 nucleotides of SEQ ID NO: 15 or at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, or at least 15, at least 16, at least 17, or at least 18 nucleotides of SEQ ID NO: 31, wherein the third probe is less than 50, less than 45, less than 40, less than 35, or less than 30 nucleotides long. In some embodiments, the fourth probe comprises at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, or at least 15, at least 16, at least 17, or at least 18 nucleotides of SEQ ID NO: 28 or at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, or at least 15, at least 16, at least 17, or at least 18 nucleotides of SEQ ID NO: 40, wherein the fourth probe is less than 50, less than 45, less than 40, less than 35, or less than 30 nucleotides long. In some embodiments, each bladder cancer marker probe comprises a dye, and wherein each dye is detectably different from the other three labels. In some embodiments, each bladder cancer marker probe comprises a fluorescent dye and a quencher molecule.

(25) In some embodiments, a composition further comprises an endogenous control primer pair for detecting an endogenous control. In some embodiments, the endogenous control is selected from ABL, GUSB, GAPDH, TUBB, and UPK1. In some embodiments, the endogenous control is ABL. In some embodiments, the endogenous control primer pair comprises a first primer comprising SEQ ID NO: 8 and a second primer comprising SEQ ID NO: 9, wherein each primer is less than 50, less than 45, less than 40, less than 35, or less than 30 nucleotides long. In some embodiments, the endogenous control primer pair comprises a first primer comprising SEQ ID NO: 41 and a second primer comprising SEQ ID NO: 42, wherein each primer is less than 50, less than 45, less than 40, less than 35, or less than 30 nucleotides long. In some embodiments, the endogenous control primer pair comprises a first primer comprising at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, or at least 15, at least 16, at least 17, or at least 18 nucleotides of SEQ ID NO: 8 and a second primer comprising at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, or at least 15, at least 16, at least 17, or at least 18 nucleotides of SEQ ID NO: 9, wherein each primer is less than 50, less than 45, less than 40, less than 35, or less than 30 nucleotides long. In some embodiments, the endogenous control primer pair comprises a first primer comprising at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, or at least 15, at least 16, at least 17, or at least 18 nucleotides of SEQ ID NO: 41 and a second primer comprising at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, or at least 15, at least 16, at least 17, or at least 18 nucleotides of SEQ ID NO: 42, wherein each primer is less than 50, less than 45, less than 40, less than 35, or less than 30 nucleotides long.

(26) In some embodiments, a composition further comprises an endogenous control probe for detecting an endogenous control amplicon. In some embodiments, the endogenous control is selected from ABL, GUSB, GAPDH, TUBB, and UPK1. In some embodiments, the endogenous control is ABL. In some embodiments, the endogenous control probe comprises at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, or at least 15, at least 16, at least 17, or at least 18 nucleotides of SEQ ID NO: 10, 11, 12, or 43, wherein the endogenous control probe is less than 50, less than 45, less than 40, less than 35, or less than 30 nucleotides long. In some embodiments, the endogenous control probe comprises a dye that is detectably different from the dyes of the bladder cancer marker probes.

(27) In some embodiments, a composition is a lyophilized composition. In some embodiments, the composition is a solution. In some embodiments, the composition further comprises urothelial cells. In some embodiments, the urothelial cells are from a urine sample.

(28) Further embodiments and details of the inventions are described below.

Description

6. DETAILED DESCRIPTION

6.1. Definitions

(1) To facilitate an understanding of the present invention, a number of terms and phrases are defined below:

(2) As used herein, the terms “detect”, “detecting” or “detection” may describe either the general act of discovering or discerning or the specific observation of a detectably labeled composition.

(3) As used herein, the term “detectably different” refers to a set of labels (such as dyes) that can be detected and distinguished simultaneously.

(4) As used herein, the terms “patient” and “subject” are used interchangeably to refer to a human. In some embodiments, the methods described herein may be used on samples from non-human animals.

(5) As used herein, “bladder cancer” is a tumor, such as a transitional cell carcinoma, arising from the lining of the bladder, and includes low grade and high grade bladder cancers, as well as metastatic bladder cancer. “Low grade bladder cancer” refers to superficial tumors that project into the interior of the bladder cavity. Low grade bladder cancers have a high rate of recurrence. “High grade bladder cancer” refers to a fast-growing and/or invasive tumor that invades the bladder wall. High grade bladder cancers have the potential to spread (i.e., metastasize) to other areas of the body. “Metastatic bladder cancer” refers to invasive bladder cancer that has spread to one or more locations in the body beyond the bladder.

(6) As used herein, the terms “oligonucleotide,” “polynucleotide,” “nucleic acid molecule,” and the like, refer to nucleic acid-containing molecules, including but not limited to, DNA or RNA. The term encompasses sequences that include any of the known base analogs of DNA and RNA including, but not limited to, 4-acetylcytosine, 8-hydroxy-N6-methyladenosine, aziridinylcytosine, pseudoisocytosine, 5-(carboxyhydroxymethyl) uracil, 5-fluorouracil, 5-bromouracil, 5-carboxymethylaminomethyl-2-thiouracil, 5-carboxymethyl-aminomethyluracil, dihydrouracil, inosine, N6-isopentenyladenine, 1-methyladenine, 1-methylpseudouracil, 1-methylguanine, 1-methylinosine, 2,2-dimethylguanine, 2-methyladenine, 2-methylguanine, 3-methylcytosine, 5-methylcytosine, N6-methyladenine, 7-methylguanine, 5-methylaminomethyluracil, 5-methoxyaminomethyl-2-thiouracil, beta-D-mannosylqueosine, 5'-methoxycarbonylmethyluracil, 5-methoxyuracil, 2-methylthio-N6-isopentenyladenine, uracil-5-oxyacetic acid methylester, uracil-5-oxyacetic acid, oxybutoxosine, pseudouracil, queosine, 2-thiocytosine, 5-methyl-2-thiouracil, 2-thiouracil, 4-thiouracil, 5-methyluracil, N-uracil-5-oxyacetic acid methylester, uracil-5-oxyacetic acid, pseudouracil, queosine, 2-thiocytosine, and 2,6-diaminopurine.

(7) As used herein, the term “oligonucleotide,” refers to a single-stranded polynucleotide having fewer than 500 nucleotides. In some embodiments, an oligonucleotide is 8 to 200, 8 to 100, 12 to 200, 12 to 100, 12 to 75, or 12 to 50 nucleotides long. Oligonucleotides may be referred to by their length, for example, a 24 residue oligonucleotide may be referred to as a “24-mer.”

(8) As used herein, the term “complementary” to a target RNA (or target region thereof), and the percentage of “complementarity” of the probe sequence to that of the target RNA sequence is the percentage “identity” to the sequence of target RNA or to the reverse complement of the sequence of the target RNA. In determining the degree of “complementarity” between probes used in the compositions described herein (or regions thereof) and a target RNA, such as those disclosed herein, the degree of “complementarity” is expressed as the percentage identity between the sequence of the probe (or region thereof) and sequence of the target RNA or the reverse complement of the sequence of the target RNA that best aligns therewith. The percentage is calculated by counting the number of aligned bases that are identical as between the 2 sequences, dividing by the total number of contiguous nucleotides in the probe, and multiplying by 100. When

the term “complementary” is used, the subject oligonucleotide is at least 90% complementary to the target molecule, unless indicated otherwise. In some embodiments, the subject oligonucleotide is at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% complementary to the target molecule.

(9) A “primer” or “probe” as used herein, refers to an oligonucleotide that comprises a region that is complementary to a sequence of at least 8 contiguous nucleotides of a target nucleic acid molecule, such as an mRNA or a DNA reverse-transcribed from an mRNA. In some embodiments, a primer or probe comprises a region that is complementary to a sequence of at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, at least 15, at least 16, at least 17, at least 18, at least 19, at least 20, at least 21, at least 22, at least 23, at least 24, at least 25, at least 26, at least 27, at least 28, at least 29, or at least 30 contiguous nucleotides of a target molecule. When a primer or probe comprises a region that is “complementary to at least x contiguous nucleotides of a target molecule,” the primer or probe is at least 95% complementary to at least x contiguous nucleotides of the target molecule. In some embodiments, the primer or probe is at least 96%, at least 97%, at least 98%, at least 99%, or 100% complementary to the target molecule.

(10) A “sample,” as used herein, includes urine samples (including samples derived from urine samples), and other types of human samples. As used herein, urine samples include, but are not limited to, whole urine, a sample comprising cells from a urine sample, a sample comprising the cell pellet isolated by centrifugation of a urine sample, a sample comprising cells isolated by filtration of a urine sample, and the like. In some embodiments, a urine sample comprises a preservative, such as a preservative that causes damage, such as lysis, of red and/or white blood cells. In some embodiments, a sample is a human sample other than a urine sample, such as a tissue sample (including a bladder tissue and/or bladder tumor sample), a blood sample (including whole blood, serum, plasma, etc.), etc. In some embodiments, a sample is a bladder washing sample.

(11) As used herein, “corticotrophin releasing hormone” or “CRH” refers to an mRNA that encodes CRH, as well as the CRH protein. In some embodiments, CRH is human CRH. Nonlimiting exemplary human CRH mRNA sequences are found at GenBank Accession No. NM_000756, and at SEQ ID NO: 1.

(12) As used herein, “insulin-like growth factor 2” or “IGF2” refers to an mRNA that encodes IGF2, as well as the IGF2 protein. In some embodiments, IGF2 is human IGF2. Nonlimiting exemplary human IGF2 mRNA sequences are shown in SEQ ID NOs: 2 to 4.

(13) As used herein, “keratin 20” or “KRT20” refers to an mRNA that encodes KRT20, as well as the KRT20 protein. In some embodiments, KRT20 is human KRT20. Nonlimiting exemplary human KRT20 mRNA sequences are found at GenBank Accession No. NM_019010, and at SEQ ID NO: 5.

(14) As used herein, “annexin A10” or “ANXA10” refers to an mRNA that encodes ANXA10, as well as the ANXA10 protein. In some embodiments, ANXA10 is human ANXA10. Nonlimiting exemplary human ANXA10 mRNA sequences are found at GenBank Accession No. NM_007193, and at SEQ ID NO: 6.

(15) An “endogenous control,” as used herein refers to a moiety that is naturally present in the sample to be used for detection, and which can be used to normalize the levels of the bladder cancer markers described herein (including, but not limited to, CRH, IGF2, KRT20 and ANXA10). Thus, an endogenous control is typically a moiety that is present at similar levels from cell to cell, and at similar levels in cells from subjects with bladder cancer and cells from subjects without bladder cancer. In some embodiments, an endogenous control is an RNA (such as an mRNA, tRNA, ribosomal RNA, etc.). Nonlimiting exemplary endogenous controls include ABL mRNA, GUSB mRNA, GAPDH mRNA, TUBB mRNA, and UPK1a mRNA. Nonlimiting exemplary human ABL mRNA sequences are found at GenBank Accession No. NM_007313, and at SEQ ID NO: 7. In some embodiments, an endogenous control is selected that can be detected in the same manner as the bladder cancer markers are detected and, in some embodiments, simultaneously with

the bladder cancer markers.

(16) An “exogenous control,” as used herein, refers to a moiety that is added to a sample to be used for detection. An exogenous control is typically selected that is not expected to be present in the sample to be used for detection, or is present at very low levels in the sample such that the amount of the moiety naturally present in the sample is either undetectable or is detectable at a much lower level than the amount added to the sample as an exogenous control. In some embodiments, an exogenous control comprises a nucleotide sequence that is not expected to be present in the sample type used for detection of the bladder cancer markers. In some embodiments, an exogenous control comprises a nucleotide sequence that is not known to be present in the species from whom the sample is taken. In some embodiments, an exogenous control comprises a nucleotide sequence from a different species than the subject from whom the sample was taken. In some embodiments, an exogenous control comprises a nucleotide sequence that is not known to be present in any species. In some embodiments, an exogenous control is selected that can be detected in the same manner as the bladder cancer markers are detected and, in some embodiments, simultaneously with the bladder cancer markers. In some embodiments, an exogenous control is an RNA. In some embodiments, an RNA is an Armored RNA®, which comprises RNA packaged in a bacteriophage protective coat. See, e.g., WalkerPeach et al., (*Clin. Chem.* 45:12: 2079-2085 (1999)).

(17) In the sequences herein, “U” and “T” are used interchangeably, such that both letters indicate a uracil or thymine at that position. One skilled in the art will understand from the context and/or intended use whether a uracil or thymine is intended and/or should be used at that position in the sequence. For example, one skilled in the art would understand that native RNA molecules typically include uracil, while native DNA molecules typically include thymine. Thus, where an RNA sequence includes “T”, one skilled in the art would understand that that position in the native RNA is likely a uracil.

(18) In the present disclosure, “a sequence selected from” encompasses both “one sequence selected from” and “one or more sequences selected from.” Thus, when “a sequence selected from” is used, it is to be understood that one, or more than one, of the listed sequences may be chosen.

(19) In the present disclosure, a method that comprises detecting a “a set of bladder cancer markers consisting of . . .” involves detection of only the bladder cancer markers of the set, and not any further bladder cancer markers. The method may comprise additional components or steps, however, such as detecting endogenous and/or exogenous controls. Similarly, a method or composition that comprises “a set of bladder cancer marker primer pairs” and/or “a set of bladder cancer marker probes” can include primer pairs and/or probes for only the bladder cancer markers of the set, and not for any other bladder cancer markers. The method or composition may comprise additional components, however, such as one or more endogenous control primer pairs and/or one or more exogenous control primer pairs.

6.2. Detecting Bladder Cancer

(20) The present inventors have developed an assay for detecting bladder cancer that involves detecting just four markers, CRH, IGF2, KRT20 and ANXA10 (plus one or two controls). The presently described assays have several advantages over existing and previously described diagnostics for bladder cancer. For example, the present assays do not rely on cytology, which can be costly, and requires trained cytologists for accurate interpretation of results. Instead, the present assays rely on the polymerase chain reaction (PCR), and can be carried out in a substantially automated manner, for example, using the GeneXpert® system (Cepheid, Sunnyvale, CA). Mengual et al., *Clin Cancer Res* (2010) 16: 2624-2633, recently described a “12+2 gene expression signature,” which detects 12 to 14 genes in an array format. Surprisingly, the present inventors have found that a much smaller signature, of just four markers (plus one or two controls), is sufficient to provide equivalent or better sensitivity and/or specificity over existing diagnostic assays for bladder cancer, including the Mengual et al. assay. Further, because the present assays use just four markers (plus one or two controls), they can be carried out in a single reaction mixture, using 4 to 6

detectably different dyes, such as fluorescent dyes. The present assays can be completed in under 3 hours, and in some embodiments, under 2 hours, using an automated system, for example, the GeneXpert® system. Existing tests can require several days for a laboratory to complete and send results. In addition, the present assay can be carried out on much smaller volumes of urine (in some embodiments, 5 ml or less). Thus, the present assays, which rely on PCR of just four markers (plus one or two controls) rather than cytology, allows for a fast, one-pot reaction for diagnosis of bladder cancer, which in many instances can be carried out at the point of care using an automated system such as GeneXpert®.

6.2.1. General Methods

(21) Compositions and methods for detecting bladder cancer are provided. In some embodiments, compositions and methods for detecting low grade bladder cancer are provided. In some embodiments, compositions and methods of detecting high grade bladder cancer are provided. In some embodiments, compositions and methods for monitoring the recurrence of bladder cancer are provided.

(22) In some embodiments, a method of detecting bladder cancer comprises detecting the levels of CRH, IGF2, KRT20 and ANXA10. In some embodiments, a method of detecting bladder cancer comprises detecting the levels of a set of markers consisting of CRH, IGF2, KRT20 and ANXA10. In some embodiments, a method of detecting bladder cancer comprises detecting the levels of CRH, IGF2, KRT20 and ANXA10, and at least one endogenous control. In some embodiments, a method of detecting bladder cancer comprises detecting the levels of a set of markers consisting of CRH, IGF2, KRT20 and ANXA10, and at least one endogenous control. In some embodiments, a method of detecting bladder cancer comprises detecting the levels of CRH, IGF2, KRT20 and ANXA10, and at least one exogenous control. In some embodiments, a method of detecting bladder cancer comprises detecting the levels of a set of markers consisting of CRH, IGF2, KRT20 and ANXA10, and at least one exogenous control. In some embodiments, a method of detecting bladder cancer comprises detecting the levels of CRH, IGF2, KRT20 and ANXA10, at least one endogenous control, and at least one exogenous control. In some embodiments, a method of detecting bladder cancer comprises detecting the levels of a set of markers consisting of CRH, IGF2, KRT20 and ANXA10, at least one endogenous control, and at least one exogenous control.

(23) In some embodiments, a method of detecting bladder cancer comprises detecting the levels of CRH, IGF2, KRT20 and ANXA10 mRNA. In some embodiments, a method of detecting bladder cancer comprises detecting the levels of a set of markers consisting of CRH, IGF2, KRT20 and ANXA10 mRNA. In some embodiments, a method of detecting bladder cancer comprises detecting the levels of CRH, IGF2, KRT20 and ANXA10 mRNA, and at least one endogenous control RNA. In some embodiments, a method of detecting bladder cancer comprises detecting the levels of a set of markers consisting of CRH, IGF2, KRT20 and ANXA10 mRNA, and at least one endogenous control RNA. In some embodiments, a method of detecting bladder cancer comprises detecting the levels of CRH, IGF2, KRT20 and ANXA10 mRNA, and at least one exogenous control RNA. In some embodiments, a method of detecting bladder cancer comprises detecting the levels of a set of markers consisting of CRH, IGF2, KRT20 and ANXA10 mRNA, and at least one exogenous control RNA. In some embodiments, a method of detecting bladder cancer comprises detecting the levels of CRH, IGF2, KRT20 and ANXA10 mRNA, at least one endogenous control RNA, and at least one exogenous control RNA. In some embodiments, a method of detecting bladder cancer comprises detecting the levels of a set of markers consisting of CRH, IGF2, KRT20 and ANXA10 mRNA, at least one endogenous control RNA, and at least one exogenous control RNA.

(24) In the present disclosure, the term “target RNA” is used for convenience to refer to CRH, IGF2, KRT20 and ANXA10 mRNAs and also to other target RNAs, such as exogenous and/or endogenous control RNAs. Thus, it is to be understood that when a discussion is presented in terms of a target RNA, that discussion is specifically intended to encompass CRH, IGF2, KRT20 and ANXA10 mRNAs, and/or other target RNAs.

(25) In some embodiments, the level of one or more target RNAs is detected in a urine sample. In some embodiments, the level of one or more target RNAs is determined in a urine sample that has been preserved in a manner that causes damage, such as lysis, to red blood cells and/or white blood cells. In some embodiments, the level of one or more target RNAs is detected in urothelial cells isolated from a urine sample, either with or without preservative treatment. In some embodiments, the urothelial cells are isolated by filtration.

(26) In some embodiments, detection of an elevated level of one or more target RNAs selected from CRH, IGF2, KRT20 and ANXA10 in a sample from a subject indicates the presence of bladder cancer in the subject. In some embodiments, the detecting is done quantitatively. In other embodiments, the detecting is done qualitatively. In some embodiments, detecting a target RNA comprises forming a complex comprising a polynucleotide and a nucleic acid selected from a target RNA, a DNA amplicon of a target RNA, and a complement of a target RNA. In some embodiments, detecting a target RNA comprises RT-PCR. In some embodiments, detecting a target RNA comprises quantitative RT-PCT. In some embodiments, the level of the target RNA is compared to a normal or control level of the target RNA.

(27) In some embodiments, the levels of target RNAs, such as CRH, IGF2, KRT20 and ANXA10 mRNA, can be measured in samples collected at one or more times from a patient to monitor the status or progression of bladder cancer in the patient. In some embodiments, a patient with a history of bladder cancer, such as a history of low grade bladder cancer or a history of high grade bladder cancer, is monitored by detecting the levels of CRH, IGF2, KRT20 and ANXA10 mRNA at regular or semi-regular intervals. In some such embodiments, the patient is monitored by detecting the levels of the target RNAs at least once per month, at least once every two months, at least once every three months, at least once every four months, at least once every five months, at least once every six months, at least once every nine months, at least once per year, or at least once every two years.

(28) In some embodiments, a sample to be tested is a urine sample (such as a voided urine sample), or is derived from a urine sample. In some embodiments, a preservative is added to the urine sample, for example, to damage (e.g., lyse) red and/or white blood cells present in the urine sample. By damaging or lysing red and/or white blood cells prior to isolation of urothelial cells, contamination by the red and/or white blood cells can be reduced. In some embodiments, the urine sample is centrifuged to concentrate the urothelial cells. In some embodiments, the urine sample is filtered to isolate the urothelial cells from other urine and preservative materials. In some such embodiments, the filter is part of a GeneXpert cartridge (Cepheid, Sunnyvale, CA).

(29) In some embodiments, less than 5 ml, less than 4 ml, less than 3 ml, or less than 2 ml of urine are used in the present methods. In some embodiments, the urine sample is analyzed without a centrifugation step. Thus, in some embodiments, the present methods are carried out in the absence of centrifugation. In some embodiments, a larger volume of urine may be used, and in some such embodiments, a centrifugation step may be used to concentrate the urothelial cells prior to analysis.

(30) In some embodiments, the sample to be tested is another bodily fluid, such as blood, sputum, mucus, saliva, semen, etc. In some embodiments, a sample to be tested is a blood sample. In some embodiments, the blood sample is whole blood. In some embodiments, the blood sample is a sample of blood cells. In some embodiments, the blood sample is plasma. In some embodiments, the blood sample is serum.

(31) The clinical sample to be tested is, in some embodiments, fresh (i.e., never frozen). In other embodiments, the sample is a frozen specimen. In some embodiments, the sample is a tissue sample, such as a formalin-fixed paraffin embedded sample. In some embodiments, the sample is a liquid cytology sample.

(32) In some embodiments, the methods described herein are used for early detection of bladder cancer in a sample of urothelial cells, such as those obtained from voided urine. In some embodiments, the methods described herein are used for monitoring for recurrence of bladder

cancer using a sample of urothelial cells, such as those obtained from voided urine.

(33) In some embodiments, the sample to be tested is obtained from an individual who has one or more of the following risk factors: history of smoking, hematuria, history of bladder or other cancers, and exposure to known carcinogens such as benzene. In some embodiments, the sample is obtained from an individual who has diagnostic signs or clinical symptoms that may be associated with bladder cancer, such as blood in the urine, frequent urination, urinary urgency, incontinence, difficulty urinating, abdominal pain, unexplained weight loss and/or loss of appetite. In some embodiments, the sample to be tested is obtained from an individual who has previously been diagnosed with low grade or high grade bladder cancer. In some such embodiments, the individual is monitored for recurrence of bladder cancer.

(34) Bladder cancer can be divided into stages, which indicate the growth pattern of the primary tumor. Table A shows the stages of bladder cancer according to the American Joint Committee on Cancer (AJCC). The stages shown cover only the “T” portion of the “TNM” system. The “T” portion refers to the primary tumor, while “N” refers to spread of the cancer to the lymph nodes, and “M” refers to whether the cancer has metastasized to distant sites.

(35) TABLE-US-00001 TABLE A Bladder cancer stages

Stage	Description
T0	No evidence of primary tumor
Ta	Non-invasive papillary carcinoma
Tis/	Non-invasive flat carcinoma (carcinoma in situ)
CIS	
T1	Tumor has grown from the lining of the bladder into the connective tissue, but has not grown into the muscle layer of the bladder
T2	Tumor has grown into the muscle layer
T2a	Tumor has grown into the inner half of the muscle layer
T2b	Tumor has grown into the outer half of the muscle layer
T3	Tumor has grown through the muscle layer of the bladder and into the fatty tissue that surrounds it
T3a	Tumor's spread to fatty tissue can only be seen under a microscope
T3b	Tumor's spread to fatty tissue can be seen on imaging tests or can be seen or felt by surgeon
T4	Tumor has spread beyond the fatty tissue to nearby organs or structures
T4a	Tumor has spread to the stroma of the prostate, or to the uterus and/or vagina
T4b	Tumor has spread to the pelvic wall or abdominal wall

(36) In some embodiments, methods described herein can be used for routine screening of healthy individuals with no risk factors. In some embodiments, methods described herein are used to screen asymptomatic individuals having one or more of the above-described risk factors.

(37) In some embodiments, the methods described herein can be used to detect low grade bladder cancer. In some embodiments, the methods described herein can be used to detect high grade bladder cancer. In some embodiments, the methods described herein detect at least 15%, at least 17%, at least 20%, at least 22%, at least 25%, at least 27%, at least 30%, at least 31%, at least 32%, at least 33%, at least 34%, or at least 35% of low grade bladder cancers. In some embodiments, the methods described herein detect at least 85%, at least 87%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% of high grade bladder cancers.

(38) In some embodiments, the methods described herein can be used to assess the effectiveness of a treatment for bladder cancer in a patient. In some embodiments, target RNA levels, such as the levels of CRH, IGF2, KRT20 or ANXA10 mRNA, are determined at various times during the treatment, and are compared to target RNA levels from an archival sample taken from the patient before the beginning of treatment. In some embodiments, target RNA levels are compared to target RNA levels from an archival normal sample taken from the patient. Ideally, target RNA levels in the normal sample evidence no aberrant changes in target RNA levels.

(39) In some embodiments, use of the levels of CRH, IGF2, KRT20 and ANXA10 mRNA for monitoring recurrence of bladder cancer is provided. In some embodiments, an elevated level of one or more, two or more, three or more, or all four mRNAs indicates that bladder cancer has recurred in the patient.

(40) In any of the embodiments described herein, RNA levels may be detected concurrently or simultaneously in the same or separate assay reactions. In some embodiments, RNA levels are

detected at different times, e.g., in serial assay reactions.

(41) In some embodiments, a method comprises detecting the level of CRH, IGF2, KRT20 and ANXA10 mRNA in a sample from a subject, wherein detection of a level of any one of the four mRNAs that is greater than a normal level of the RNA indicates the presence of bladder cancer in the subject. In some embodiments, detection of elevated levels of two, three, or four bladder cancer marker mRNAs indicates the presence of bladder cancer in the subject. In some embodiments, detection of elevated levels of at least two, at least three, or all four of the bladder cancer marker mRNAs indicates a greater risk of high grade bladder cancer.

(42) In some embodiments, a method of facilitating diagnosis of bladder cancer in a subject is provided. Such methods comprise detecting the levels of CRH, IGF2, KRT20 and ANXA10 mRNA in a sample from the subject. In some embodiments, information concerning the levels of CRH, IGF2, KRT20 and/or ANXA10 mRNA in the sample from the subject is communicated to a medical practitioner. A “medical practitioner,” as used herein, refers to an individual or entity that diagnoses and/or treats patients, such as a hospital, a clinic, a physician's office, a physician, a nurse, or an agent of any of the aforementioned entities and individuals. In some embodiments, detecting the levels of CRH, IGF2, KRT20 and ANXA10 mRNA is carried out at a laboratory that has received the subject's sample from the medical practitioner or agent of the medical practitioner. The laboratory carries out the detection by any method, including those described herein, and then communicates the results to the medical practitioner. A result is “communicated,” as used herein, when it is provided by any means to the medical practitioner. In some embodiments, such communication may be oral or written, may be by telephone, in person, by e-mail, by mail or other courier, or may be made by directly depositing the information into, e.g., a database accessible by the medical practitioner, including databases not controlled by the medical practitioner. In some embodiments, the information is maintained in electronic form. In some embodiments, the information can be stored in a memory or other computer readable medium, such as RAM, ROM, EEPROM, flash memory, computer chips, digital video discs (DVD), compact discs (CDs), hard disk drives (HDD), magnetic tape, etc.

(43) In some embodiments, methods of detecting the presence bladder cancer are provided. In some embodiments, methods of diagnosing bladder cancer are provided. In some embodiments, the method comprises obtaining a sample from a subject and providing the sample to a laboratory for detection of levels of CRH, IGF2, KRT20 and ANXA10 mRNA in the sample. In some embodiments, the method further comprises receiving a communication from the laboratory that indicates the level of at least one RNA selected from CRH, IGF2, KRT20 and ANXA10 mRNA in the sample. In some embodiments, bladder cancer is present if the level of any one of the four mRNAs is greater than a normal or control level of the mRNA. A “laboratory,” as used herein, is any facility that detects the levels of CRH, IGF2, KRT20 and ANXA10 mRNA in a sample by any method, including the methods described herein, and communicates the level to a medical practitioner. In some embodiments, a laboratory is under the control of a medical practitioner. In some embodiments, a laboratory is not under the control of the medical practitioner.

(44) When a laboratory communicates the level of at least one RNA selected from CRH, IGF2, KRT20 and ANXA10 to a medical practitioner, in some embodiments, the laboratory communicates a numerical value representing the level of the RNA in the sample, with or without providing a numerical value for a normal level. In some embodiments, the laboratory communicates the level of the RNA by providing a qualitative value, such as “high,” “low,” “elevated,” “decreased,” “positive” (such as “CRH positive” or “CRH and KRT20 positive”), etc. In some embodiments, the laboratory communicates a suggested diagnosis, such as “bladder cancer positive” or “positive for cancer,” and the like; or simply “cancer positive” or “cancer negative.”

(45) As used herein, when a method relates to detecting bladder cancer, determining the presence of bladder cancer, monitoring for bladder cancer, and/or diagnosing bladder cancer, the method includes activities in which the steps of the method are carried out, but the result is negative for the

presence of bladder cancer. That is, detecting, determining, monitoring, and diagnosing bladder cancer include instances of carrying out the methods that result in either positive or negative results.

(46) In some embodiments, more than one RNA is detected simultaneously in a single reaction. In some embodiments, CRH, IGF2, KRT20 and ANXA10 mRNAs are detected simultaneously in a single reaction. In some embodiments, CRH, IGF2, KRT20 and ANXA10 mRNAs and at least one endogenous control and/or at least one exogenous control are detected simultaneously in a single reaction. In some embodiments, CRH, IGF2, KRT20 and ANXA10 mRNAs, and endogenous control, and an exogenous control are detected simultaneously in a single reaction.

6.2.2. Exemplary Controls

(47) In some embodiments, a normal level (a “control”) of a target RNA, such as CRH, IGF2, KRT20 or ANXA10 mRNA, can be determined as an average level or range that is characteristic of normal urothelial cells or other reference material, against which the level measured in the sample can be compared. The determined average or range of a target RNA in normal subjects can be used as a benchmark for detecting above-normal levels of the target RNA that are indicative of bladder cancer. In some embodiments, normal levels of a target RNA can be determined using individual or pooled RNA-containing samples from one or more individuals, such as from normal urothelial cells isolated from urine of healthy individuals.

(48) In some embodiments, determining a normal level of a target RNA, such as CRH, IGF2, KRT20 or ANXA10 mRNA, comprises detecting a complex comprising a polynucleotide for detection hybridized to a nucleic acid selected from a target RNA, a DNA amplicon of the target RNA, and a complement of the target RNA. That is, in some embodiments, a normal level can be determined by detecting a DNA amplicon of the target RNA, or a complement of the target RNA rather than the target RNA itself. In some embodiments, a normal level of such a complex is determined and used as a control. The normal level of the complex, in some embodiments, correlates to the normal level of the target RNA. Thus, when a normal level of a target is discussed herein, that level can, in some embodiments, be determined by detecting such a complex.

(49) In some embodiments, a normal level of a target RNA is, or has been, determined by the same method as the level of the target RNA from a patient sample. In some such embodiments, the method is RT-PCR (such as real-time RT-PCR, quantitative RT-PCR, etc.).

(50) In some embodiments, a control comprises RNA from cells of a single individual, e.g., from normal urothelial cells isolated from urine of a healthy individual. In some embodiments, a control comprises RNA from blood, such as whole blood or serum, of a single individual. In some embodiments, a control comprises RNA from a pool of cells from multiple individuals. In some embodiments, a control comprises RNA from a pool of urine from multiple individuals. In some embodiments, a control comprises commercially-available human RNA (see, for example, Ambion). In some embodiments, a normal level or normal range has already been predetermined prior to testing a sample for an elevated level.

(51) In some embodiments, the normal level of a target RNA can be determined from one or more continuous cell lines, typically cell lines previously shown to have levels of RNAs that approximate the levels in normal urothelial cells.

(52) In some embodiments, quantitation of target RNA levels requires assumptions to be made about the total RNA per cell and the extent of sample loss during sample preparation. In order to correct for differences between different samples or between samples that are prepared under different conditions, the quantities of target RNAs in some embodiments are normalized to the levels of at least one endogenous control and/or at least one exogenous control.

(53) In some embodiments, a control RNA is an endogenous control RNA. An endogenous control RNA may be any RNA suitable for the purpose, for example, RNAs that are present at approximately constant levels from cell to cell and in urothelial cells from both bladder cancer and non-bladder cancer patients. Nonlimiting exemplary endogenous control RNAs include ABL,

GUSB, GAPDH, TUBB, and UPK1a. In some embodiments, one endogenous control is used for normalization. In some embodiments, more than one endogenous control is used for normalization. (54) In some embodiments, the level of a target RNA, such as CRH, IGF2, KRT20 or ANXA10 mRNA, is normalized to an endogenous control RNA. Normalization may comprise, for example, determination of the difference of the level of the target RNA to the level of the endogenous control RNA. In some such embodiments, the level of the RNAs are represented by a Ct value obtained from quantitative PCR. In some such embodiments, the difference is expressed as ΔCt . ΔCt may be calculated as $Ct[\text{target RNA}] - Ct[\text{endogenous control}]$ or $Ct[\text{endogenous control}] - Ct[\text{target RNA}]$. In certain embodiments, $\Delta Ct = Ct[\text{endogenous control}] - Ct[\text{marker}]$. In some embodiments, a threshold ΔCt value is set, above or below which bladder cancer is indicated. In some such embodiments, the ΔCt threshold is set as the ΔCt value below which 95% of normal samples are correctly characterized. In some such embodiments, a ΔCt value that is higher than the threshold ΔCt value is indicative of bladder cancer.

(55) In some embodiments, linear discriminant analysis (LDA) is used, for example, to combine two or more of the markers into a single combined scale. In some such embodiments, a single threshold value is used for the markers included in the LDA.

(56) In some embodiments, a control RNA is an exogenous control RNA. In some such embodiments, the exogenous control RNA is an Armored RNA®, which is protected by a bacteriophage coat. An exogenous control RNA may, in some embodiments, be used to determine if the detection assay reaction has failed, and therefore the results are not meaningful. For example, if an exogenous control RNA is not amplified in the assay reaction, then a negative result for the target RNAs is likely not meaningful because the levels reflect the reaction failing rather than the target RNA levels being low. Reaction failure can occur for any number of reasons, including, but not limited to, the presence of a reaction inhibitor in the sample (an “inhibitory sample”), compromised reagents, the presence of an RNase, etc. An exogenous RNA control may be added at any stage of the sample collection and analysis. For example, in some embodiments, the exogenous control RNA is added to the sample at the time preservative is added, is added to the sample when it is received by the diagnostic laboratory, is added to the sample immediately prior to analysis, or is added to the sample during analysis (as a nonlimiting example, during or after lysis of the urothelial cells but before addition of the amplification reagents).

(57) In some embodiments, the level of a target RNA, such as CRH, IGF2, KRT20 or ANXA10 mRNA, is compared to a reference level, e.g., from a confirmed bladder cancer. In some such embodiments, a similar level of a target RNA relative to the reference sample indicates bladder cancer.

(58) In some embodiments, a level of a target RNA, such as CRH, IGF2, KRT20 or ANXA10 mRNA, that is at least about 20%, 30%, 40%, 50%, 60%, 70%, 80%, or 90% greater than a normal level of the respective target RNA indicates the presence of bladder cancer. In some embodiments, a level of a target RNA, such as CRH, IGF2, KRT20 or ANXA10 mRNA, that is at least about two-fold, at least about 3-fold, at least about 4-fold, at least about 5-fold, at least about 6-fold, at least about 7-fold, at least about 8-fold, at least about 9-fold, or at least about 10-fold greater than a normal level of the respective target RNA indicates the presence of bladder cancer.

(59) In some embodiments, a control level of a target RNA, such as CRH, IGF2, KRT20 or ANXA10 mRNA, is determined contemporaneously, such as in the same assay or batch of assays, as the level of the target RNA in a sample. In some embodiments, a control level of a target RNA is not determined contemporaneously as the level of the target RNA in a sample. In some such embodiments, the control level has been determined previously.

(60) In some embodiments, the level of an endogenous control and/or an exogenous control is determined contemporaneously, such as in the same assay or batch of assays, as the level of the target RNA in a sample. In some embodiments, an assay comprises reagents for determining the levels of CRH, IGF2, KRT20 and ANXA10 mRNA, and an endogenous control simultaneously in

the same assay reaction. In some embodiments, an assay comprises reagents for determining the levels of CRH, IGF2, KRT20 and ANXA10 mRNA, and an exogenous control simultaneously in the same assay reaction. In some embodiments, an assay comprises reagents for determining the levels of CRH, IGF2, KRT20, ANXA10 mRNA, an endogenous control, and an exogenous control simultaneously in the same assay reaction. In some such embodiments, for example, an assay reaction comprises primer sets for amplifying each of CRH, IGF2, KRT20 and ANXA10 mRNAs, a primer set for amplifying an endogenous control and/or a primer set for amplifying an exogenous control, and detectably different labeled probes for detecting the amplification products (such as, for example, TaqMan® probes with detectably different dyes for each different amplicon to be detected).

(61) In some embodiments, the level of a target RNA is not compared to a control level, for example, when it is known that the target RNA is present at very low levels, or not at all, in normal cells. In such embodiments, detection of a high level of the target RNA in a sample is indicative of bladder cancer.

6.2.3. Exemplary Sample Preparation

6.2.3.1. Exemplary Urine Preservatives

(62) In some embodiments, a preservative is added to the urine sample. In some embodiments, the preservative is added within one hour, two hours, three hours, or six hours of the time the urine sample was collected (e.g., voided). In some embodiments, a preservative is added to the urine sample within one hour, two hours, three hours, or six hours before the sample is analyzed by the methods described herein.

(63) In some embodiments, a preservative causes damage, such as lysis, of red blood cells and/or white blood cells, but does not damage urothelial cells. Red blood cells and/or white blood cells may be present in the urine as a result of a tumor and/or infection. In some such embodiments, adding the preservative allows for improved enrichment of the urothelial cells, for example by filtration. In some embodiments, a preservative lowers the pH of the urine sample and improves solubility of urine salts. In some such embodiments, the preservative facilitates passage of the salts through a filter in a filtration step. A desirable pH of preserved urine to be passed through a filter is between about 2.5 and 4. In some embodiments, a desirable pH of preserved urine is between about 2.7 and 3.7. In some embodiments, a desirable pH of preserved urine is between about 3 and 3.5. In some embodiments, a desirable pH of preserved urine is about 3.2.

(64) In some embodiments a preservative is added such that the urine/preservative sample comprises 0.875M to 2.625M guanidine hydrochloride, 0.25% to 0.75% N-acetyl-L-cysteine, 6.25 to 18.75 mM sodium citrate, and 0.625% to 1.875% Tween-20, and has a pH of 3 to 3.5. In some embodiments a preservative is added such that the urine/preservative sample comprises about 1.75 M guanidine hydrochloride, about 0.5% N-acetyl-L-cysteine, about 12.5 mM sodium citrate, and about 1.25% Tween-20, and has a pH of about 3.2.

(65) A nonlimiting exemplary commercial preservative is PreservCyt (Hologic, Bedford, MA).

6.2.3.2. Exemplary Cell Enrichment

(66) In some embodiments, urothelial cells are enriched by centrifugation. In some such embodiments, the cell pellet is resuspended in the supernatant and/or a preservative. Resuspension of the cell pellet can be used to adjust the concentration of cells in solution. The resuspended cell pellet may be used (for example, with lysis) in the methods described herein, or may be subject to an additional enrichment step, such as filtration.

(67) In some embodiments, urothelial cells are enriched by filtration. Nonlimiting exemplary filter pore sizes that may be suitable for capturing urothelial cells include 0.8 μm , 2 μm , 8 μm , and 10 μm . In some embodiments, a filter pore size is selected that allows pass-through of red blood cells and/or white blood cells, while retaining most urothelial cells. In some embodiments, a filter is located within a GeneXpert cartridge designed for carrying out a bladder cancer diagnostic assay described herein.

6.2.3.3. Exemplary mRNA Preparation

(68) Target RNA can be prepared by any appropriate method. Total RNA can be isolated by any method, including, but not limited to, the protocols set forth in Wilkinson, M. (1988) Nucl. Acids Res. 16(22): 10,933; and Wilkinson, M. (1988) Nucl. Acids Res. 16(22): 10934, or by using commercially-available kits or reagents, such as the TRIzol® reagent (Invitrogen), Total RNA Extraction Kit (iNtRON Biotechnology), Total RNA Purification Kit (Norgen Biotek Corp.), RNAqueous™ (Ambion), MagMAX™ (Ambion), RecoverAll™ (Ambion), RNeasy (Qiagen), etc.

(69) In some embodiments, RNA levels are measured in a sample in which RNA has not first been purified from the cells. In some such embodiments, the cells are subject to a lysis step to release the RNA. Nonlimiting exemplary lysis methods include sonication (for example, for 2-15 seconds, 8-18 μm at 36 kHz); chemical lysis, for example, using a detergent; and various commercially available lysis reagents (such as RNeasy lysis buffer, Qiagen). In some embodiments, RNA levels are measured in a sample in which RNA has been isolated.

(70) In some embodiments, RNA is modified before a target RNA, such as CRH, IGF2, KRT20 or ANXA10 mRNA, is detected. In some embodiments, all of the RNA in the sample is modified. In some embodiments, just the particular target RNAs to be analyzed are modified, e.g., in a sequence-specific manner. In some embodiments, RNA is reverse transcribed. In some such embodiments, RNA is reverse transcribed using MMLV reverse transcriptase. Nonlimiting exemplary conditions for reverse transcribing RNA using MMLV reverse transcriptase include incubation from 5 to 20 minutes at 40° ° C. to 50° ° C.

(71) When a target RNA is reverse transcribed, a DNA complement of the target RNA is formed. In some embodiments, the complement of a target RNA is detected rather than a target RNA itself (or a DNA copy of the RNA itself). Thus, when the methods discussed herein indicate that a target RNA is detected, or the level of a target RNA is determined, such detection or determination may be carried out on a complement of a target RNA instead of, or in addition to, the target RNA itself. In some embodiments, when the complement of a target RNA is detected rather than the target RNA, a polynucleotide for detection is used that is complementary to the complement of the target RNA. In some such embodiments, a polynucleotide for detection comprises at least a portion that is identical in sequence to the target RNA, although it may contain thymidine in place of uridine, and/or comprise other modified nucleotides.

6.2.4. Exemplary Analytical Methods

(72) As described above, methods are presented for detecting bladder cancer. The methods comprise detecting a panel of bladder cancer markers consisting of CRH, IGF2, KRT20 and ANXA10, and optionally including at least one endogenous control and/or at least one exogenous control. In some embodiments, detection of an elevated level of one of the four bladder cancer markers indicates the presence of bladder cancer. In some embodiments, detection of an elevated level of two, three, or all four of the bladder cancer markers indicates the presence of bladder cancer. In some embodiments, the bladder cancer is low grade bladder cancer. In some embodiments, the bladder cancer is high grade bladder cancer. In some embodiments, the bladder cancer is a recurrence of bladder cancer in a patient with a history of bladder cancer.

(73) Any analytical procedure capable of permitting specific and quantifiable (or semi-quantifiable) detection of a target RNA, such as CRH, IGF2, KRT20 and ANXA10 mRNAs, may be used in the methods herein presented. Such analytical procedures include, but are not limited to, RT-PCR methods, and other methods known to those skilled in the art.

(74) In some embodiments, the method of detecting a target RNA, such as CRH, IGF2, KRT20 or ANXA10 mRNA, comprises amplifying cDNA complementary to the target RNA. Such amplification can be accomplished by any method. Exemplary methods include, but are not limited to, real time PCR, endpoint PCR, and amplification using T7 polymerase from a T7 promoter annealed to a cDNA, such as provided by the SenseAmp Plus™ Kit available at Implen, Germany.

(75) When a target RNA or a cDNA complementary to a target RNA is amplified, in some

embodiments, a DNA amplicon of the target RNA is formed. A DNA amplicon may be single stranded or double-stranded. In some embodiments, when a DNA amplicon is single-stranded, the sequence of the DNA amplicon is related to the target RNA in either the sense or antisense orientation. In some embodiments, a DNA amplicon of a target RNA is detected rather than the target RNA itself. Thus, when the methods discussed herein indicate that a target RNA is detected, or the level of a target RNA is determined, such detection or determination may be carried out on a DNA amplicon of the target RNA instead of, or in addition to, the target RNA itself. In some embodiments, when the DNA amplicon of the target RNA is detected rather than the target RNA, a polynucleotide for detection is used that is complementary to the complement of the target RNA. In some embodiments, when the DNA amplicon of the target RNA is detected rather than the target RNA, a polynucleotide for detection is used that is complementary to the target RNA. Further, in some embodiments, multiple polynucleotides for detection may be used, and some polynucleotides may be complementary to the target RNA and some polynucleotides may be complementary to the complement of the target RNA.

(76) In some embodiments, the method of detecting one or more target RNAs, such as CRH, IGF2, KRT20 or ANXA10 mRNA, comprises RT-PCR, as described below. In some embodiments, detecting one or more target RNAs comprises real-time monitoring of an RT-PCR reaction, which can be accomplished by any method. Such methods include, but are not limited to, the use of TaqMan®, Molecular beacon, or Scorpion probes (i.e., energy transfer (ET) probes, such as FRET probes) and the use of intercalating dyes, such as SYBR green, EvaGreen, thiazole orange, YO-PRO, TO-PRO, etc.

(77) Nonlimiting exemplary conditions for amplifying cDNA that has been reverse transcribed from the target RNAs are as follows. An exemplary cycle comprises an initial denaturation at 90° C. to 100° C. for 2 to 5 minutes, followed by cycling that comprises denaturation at 90° C. to 100° C. for 1 to 10 seconds, annealing at 60° C. to 70° C. for 10 to 30 seconds, and extension at 60° C. to 75° C. for 10 to 40 seconds. In some embodiments, for the first cycle following the initial denaturation step, the cycle denaturation step is omitted. In some embodiments, Taq polymerase is used for amplification. In some embodiments, the cycle is carried out at least 10 times, at least 15 times, at least 20 times, at least 25 times, at least 30 times, at least 35 times, or at least 45 times. In some such embodiments, Taq is used with a hot start function. In some embodiments, the amplification reaction occurs in a GeneXpert cartridge, and amplification of the four bladder cancer marker target RNAs occurs in the same reaction. In some embodiments, detection of CRH, IGF2, KRT20 and ANXA10 mRNAs occurs in less than 3 hours, less than 2.5 hours, or less than 2 hours, from initial denaturation through the last extension.

(78) In some embodiments, detection of a target RNA comprises forming a complex comprising a polynucleotide that is complementary to a target RNA or to a complement thereof, and a nucleic acid selected from the target RNA, a DNA amplicon of the target RNA, and a complement of the target RNA. Thus, in some embodiments, the polynucleotide forms a complex with a target RNA. In some embodiments, the polynucleotide forms a complex with a complement of the target RNA, such as a cDNA that has been reverse transcribed from the target RNA. In some embodiments, the polynucleotide forms a complex with a DNA amplicon of the target RNA. When a double-stranded DNA amplicon is part of a complex, as used herein, the complex may comprise one or both strands of the DNA amplicon. Thus, in some embodiments, a complex comprises only one strand of the DNA amplicon. In some embodiments, a complex is a triplex and comprises the polynucleotide and both strands of the DNA amplicon. In some embodiments, the complex is formed by hybridization between the polynucleotide and the target RNA, complement of the target RNA, or DNA amplicon of the target RNA. The polynucleotide, in some embodiments, is a primer or probe.

(79) In some embodiments, a method comprises detecting the complex. In some embodiments, the complex does not have to be associated at the time of detection. That is, in some embodiments, a complex is formed, the complex is then dissociated or destroyed in some manner, and components

from the complex are detected. An example of such a system is a TaqMan® assay. In some embodiments, when the polynucleotide is a primer, detection of the complex may comprise amplification of the target RNA, a complement of the target RNA, or a DNA amplicon of a target RNA.

(80) In some embodiments the analytical method used for detecting at least one target RNA in the methods set forth herein includes real-time quantitative RT-PCR. In some embodiments, the analytical method used for detecting at least one target RNA includes the use of a TaqMan® probe. The assay uses energy transfer (“ET”), such as fluorescence resonance energy transfer (“FRET”), to detect and quantitate the synthesized PCR product. Typically, the TaqMan® probe comprises a fluorescent dye molecule coupled to the 5'-end and a quencher molecule coupled to the 3'-end, such that the dye and the quencher are in close proximity, allowing the quencher to suppress the fluorescence signal of the dye via FRET. When the polymerase replicates the chimeric amplicon template to which the TaqMan® probe is bound, the 5'-nuclease of the polymerase cleaves the probe, decoupling the dye and the quencher so that the dye signal (such as fluorescence) is detected. Signal (such as fluorescence) increases with each RT-PCR cycle proportionally to the amount of probe that is cleaved.

(81) In some embodiments, quantitation of the results of real-time RT-PCR assays is done by constructing a standard curve from a nucleic acid of known concentration and then extrapolating quantitative information for target RNAs of unknown concentration. In some embodiments, the nucleic acid used for generating a standard curve is an RNA (for example, an endogenous control, or an exogenous control). In some embodiments, the nucleic acid used for generating a standard curve is a purified double-stranded plasmid DNA or a single-stranded DNA generated in vitro.

(82) In some embodiments, where the amplification efficiencies of the target nucleic acids and the endogenous reference are approximately equal, quantitation is accomplished by the comparative Ct (cycle threshold, e.g., the number of PCR cycles required for the fluorescence signal to rise above background) method. Ct values are inversely proportional to the amount of nucleic acid target in a sample. In some embodiments, Ct values of a target RNA can be compared with a control or calibrator, such as an exogenous control RNA. In some embodiments, the Ct values of the exogenous control and the target RNA are normalized to an appropriate endogenous control. Nonlimiting exemplary endogenous controls are discussed herein.

(83) In some embodiments, a threshold Ct (or a “cutoff Ct”) value for a target RNA, below which bladder cancer is indicated, has previously been determined. In such embodiments, a control sample may not be assayed concurrently with the test sample. In some embodiments, as discussed herein, a Δ Ct threshold value is determined, above which bladder cancer is indicated, has previously been determined.

(84) In addition to the TaqMan® assays, other real-time RT-PCR chemistries useful for detecting and quantitating PCR products in the methods presented herein include, but are not limited to, Molecular Beacons, Scorpion probes and intercalating dyes, such as SYBR Green, EvaGreen, thiazole orange, YO-PRO, TO-PRO, etc., which are discussed below.

(85) In various embodiments, real-time RT-PCR detection is utilized to detect, in a single multiplex reaction, all four bladder cancer markers of the panel described herein, and optionally, at least one endogenous control and/or at least one exogenous control. In some multiplex embodiments, a plurality of probes, such as TaqMan® probes, each specific for a different RNA target, is used. In some embodiments, each target RNA-specific probe is spectrally distinguishable from the other probes used in the same multiplex reaction.

(86) In some embodiments, quantitation of real-time RT PCR products is accomplished using a dye that binds to double-stranded DNA products, such as SYBR Green, EvaGreen, thiazole orange, YO-PRO, TO-PRO, etc. In some embodiments, the assay is the QuantiTect SYBR Green PCR assay from Qiagen. In this assay, total RNA is first isolated from a sample. Total RNA is subsequently poly-adenylated at the 3'-end and reverse transcribed using a universal primer with

poly-dT at the 5'-end. In some embodiments, a single reverse transcription reaction is sufficient to assay multiple target RNAs. Real-time RT-PCR is then accomplished using target RNA-specific primers and an miScript Universal Primer, which comprises a poly-dT sequence at the 5'-end. SYBR Green dye binds non-specifically to double-stranded DNA and upon excitation, emits light. In some embodiments, buffer conditions that promote highly-specific annealing of primers to the PCR template (e.g., available in the QuantiTect SYBR Green PCR Kit from Qiagen) can be used to avoid the formation of non-specific DNA duplexes and primer dimers that will bind SYBR Green and negatively affect quantitation. Thus, as PCR product accumulates, the signal from SYBR Green increases, allowing quantitation of specific products.

(87) Real-time RT-PCR is performed using any RT-PCR instrumentation available in the art. Typically, instrumentation used in real-time RT-PCR data collection and analysis comprises a thermal cycler, optics for fluorescence excitation and emission collection, and optionally a computer and data acquisition and analysis software.

(88) In some embodiments, the analytical method used in the methods described herein is a DASL® (cDNA-mediated Annealing, Selection, Extension, and Ligation) Assay. In some embodiments, total RNA is isolated from a sample to be analyzed by any method. Total RNA may then be polyadenylated (>18 A residues are added to the 3'-ends of the RNAs in the reaction mixture). The RNA is reverse transcribed using a biotin-labeled DNA primer that comprises from the 5' to the 3' end, a sequence that includes a PCR primer site and a poly-dT region that binds to the poly-dA tail of the sample RNA. The resulting biotinylated cDNA transcripts are then hybridized to a solid support via a biotin-streptavidin interaction and contacted with one or more target RNA-specific polynucleotides. The target RNA-specific polynucleotides comprise, from the 5'-end to the 3'-end, a region comprising a PCR primer site, region comprising an address sequence, and a target RNA-specific sequence.

(89) In some DASL® embodiments, the target RNA-specific sequence comprises at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, at least 15, at least 16, at least 17, at least 18, at least 19 contiguous nucleotides having a sequence that is the same as, or complementary to, at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, at least 15, at least 16, at least 17, at least 18, at least 19 contiguous nucleotides of a bladder cancer marker target RNA, an endogenous control RNA, or an exogenous control RNA.

(90) After hybridization, the target RNA-specific polynucleotide is extended, and the extended products are then eluted from the immobilized cDNA array. A second PCR reaction using a fluorescently-labeled universal primer generates a fluorescently-labeled DNA comprising the target RNA-specific sequence. The labeled PCR products are then hybridized to a microbead array for detection and quantitation.

(91) In some embodiments, the analytical method used for detecting and quantifying the levels of the at least one target RNA in the methods described herein is a bead-based flow cytometric assay. See Lu J. et al. (2005) Nature 435:834-838, which is incorporated herein by reference in its entirety. An example of a bead-based flow cytometric assay is the xMAP® technology of Luminex, Inc. (See luminexcorp.com/technology/index.html). In some embodiments, total RNA is isolated from a sample and is then labeled with biotin. The labeled RNA is then hybridized to target RNA-specific capture probes (e.g., FlexmiR™ products sold by Luminex, Inc. at luminexcorp.com/products/assays/index.html) that are covalently bound to microbeads, each of which is labeled with 2 dyes having different fluorescence intensities. A streptavidin-bound reporter molecule (e.g., streptavidin-phycoerythrin, also known as "SAPE") is attached to the captured target RNA and the unique signal of each bead is read using flow cytometry. In some embodiments, the RNA sample is first polyadenylated, and is subsequently labeled with a biotinylated 3DNA™ dendrimer (i.e., a multiple-arm DNA with numerous biotin molecules bound thereto), using a bridging polynucleotide that is complementary to the 3'-end of the poly-dA tail of the sample RNA and to the 5'-end of the polynucleotide attached to the biotinylated dendrimer. The

streptavidin-bound reporter molecule is then attached to the biotinylated dendrimer before analysis by flow cytometry. In some embodiments, biotin-labeled RNA is first exposed to SAPE, and the RNA/SAPE complex is subsequently exposed to an anti-phycoerythrin antibody attached to a DNA dendrimer, which can be bound to as many as 900 biotin molecules. This allows multiple SAPE molecules to bind to the biotinylated dendrimer through the biotin-streptavidin interaction, thus increasing the signal from the assay.

(92) In some embodiments, the analytical method used for detecting and quantifying the levels of the at least one target RNA in the methods described herein is by gel electrophoresis and detection with labeled probes (e.g., probes labeled with a radioactive or chemiluminescent label), such as by Northern blotting. In some embodiments, total RNA is isolated from the sample, and then is size-separated by SDS polyacrylamide gel electrophoresis. The separated RNA is then blotted onto a membrane and hybridized to radiolabeled complementary probes. In some embodiments, exemplary probes contain one or more affinity-enhancing nucleotide analogs as discussed below, such as locked nucleic acid (“LNA”) analogs, which contain a bicyclic sugar moiety instead of deoxyribose or ribose sugars. See, e.g., Várallyay, E. et al. (2008) *Nature Protocols* 3(2):190-196, which is incorporated herein by reference in its entirety.

(93) In some embodiments, detection and quantification of one or more target RNAs is accomplished using microfluidic devices and single-molecule detection. In some embodiments, target RNAs in a sample of isolated total RNA are hybridized to two probes, one which is complementary to nucleic acids at the 5'-end of the target RNA and the second which is complementary to the 3'-end of the target RNA. Each probe comprises, in some embodiments, one or more affinity-enhancing nucleotide analogs, such as LNA nucleotide analogs and each is labeled with a different fluorescent dye having different fluorescence emission spectra (i.e., detectably different dyes). The sample is then flowed through a microfluidic capillary in which multiple lasers excite the fluorescent probes, such that a unique coincident burst of photons identifies a particular target RNA, and the number of particular unique coincident bursts of photons can be counted to quantify the amount of the target RNA in the sample. In some alternative embodiments, a target RNA-specific probe can be labeled with 3 or more distinct labels selected from, e.g., fluorophores, electron spin labels, etc., and then hybridized to an RNA sample.

(94) Optionally, the sample RNA is modified before hybridization. The target RNA/probe duplex is then passed through channels in a microfluidic device and that comprise detectors that record the unique signal of the 3 labels. In this way, individual molecules are detected by their unique signal and counted. See U.S. Pat. Nos. 7,402,422 and 7,351,538 to Fuchs et al., U.S. Genomics, Inc., each of which is incorporated herein by reference in its entirety.

6.2.5. Exemplary Automation and Systems

(95) In some embodiments, gene expression is detected using an automated sample handling and/or analysis platform. In some embodiments, commercially available automated analysis platforms are utilized. For example, in some embodiments, the GeneXpert® system (Cepheid, Sunnyvale, CA) is utilized.

(96) The present invention is illustrated for use with the GeneXpert system. Exemplary sample preparation and analysis methods are described below. However, the present invention is not limited to a particular detection method or analysis platform. One of skill in the art recognizes that any number of platforms and methods may be utilized.

(97) The GeneXpert® utilizes a self-contained, single use cartridge. Sample extraction, amplification, and detection may all carried out within this self-contained “laboratory in a cartridge.” (See e.g., U.S. Pat. Nos. 5,958,349, 6,403,037, 6,440,725, 6,783,736, 6,818,185; each of which is herein incorporated by reference in its entirety.)

(98) Components of the cartridge include, but are not limited to, processing chambers containing reagents, filters, and capture technologies useful to extract, purify, and amplify target nucleic acids. A valve enables fluid transfer from chamber to chamber and contain nucleic acids lysis and

filtration components. An optical window enables real-time optical detection. A reaction tube enables very rapid thermal cycling.

(99) In some embodiments, the GenXpert® system includes a plurality of modules for scalability. Each module includes a plurality of cartridges, along with sample handling and analysis components.

(100) In some embodiments, the GeneXpert® sample preparation method utilizes filtration in order to capture and concentrate cells from urine. In some embodiments, a filter pore size of 0.8 μm is utilized. This size facilitates capture of all cells in urine. In other embodiments, pore sizes of 0.5 to 10 μm , 0.5 to 5 μm , 0.8 to 10 μm , 0.8 to 5 μm , 0.8 to 2 μm , 2 to 5 μm , 2 to 10 μm , 2 to 8 μm , 5 to 8 μm , or 5 to 10 μm are utilized. Certain filters (such as 5 μm , 8 μm , and 10 μm) allow the removal of most red and white blood cells from the sample while capturing the larger urothelial cells, which are the assay target cells. In some embodiments, this sample preparation method improves assay specificity by removing white blood cells that may be present due to infection or inflammation. In some instances, sample preparation methods such as centrifugation of whole urine followed by RNA isolation from the urine pellet do not allow for removal of white blood cells. In some embodiments, the efficiency of cell capture by filtration is higher compared to centrifugation, and may provide more consistent results.

(101) After the cells from the urine are captured on the filter, in some embodiments, they are washed and then lysed using sonication (2-15 seconds, 8-16 μm at 36 kHz). The cell lysate is then collected and used to reconstitute the RT-PCR reagents, which are present in the cartridge as lyophilized particles.

(102) In some embodiments, RT-PCR is used to amplify and analyze the presence or expression levels of the bladder cancer markers. In some embodiments, the reverse transcription uses MMLV RT enzyme and an incubation of 5 to 20 minutes at 40° C. to 50° C. In some embodiments, the PCR uses Taq polymerase with hot start function, such as AptaTaq (Roche). In some embodiments, the initial denaturation is at 90° C. to 100° C. for 2 to 5 minutes; the cycling denaturation temperature is 90° C. to 100° C. for 1 to 10 seconds; the cycling anneal temperature is 60° C. to 70° C. for 10 to 30 seconds; and the cycling extend temperature is 60° C. to 75° C. for 10 to 40 seconds; and up to 50 cycles are performed.

(103) The present invention is not limited to particular primer and/or probe sequences. Exemplary amplification primers and detection probes are described in the Examples.

(104) In some embodiments, an off-line centrifugation is used to improve assay results with samples with low cellular content. The sample, with or without the preservative added, is centrifuged and the supernatant removed. The pellet is then resuspended in a smaller volume of either supernatant or the preservative. The resuspended pellet is then added to a GeneXpert® cartridge as previously described.

6.2.6. Exemplary Data Analysis

(105) In some embodiments, a computer-based analysis program is used to translate the raw data generated by the detection assay (e.g., the expression level of the bladder cancer markers described herein) into data of predictive value for a clinician. The clinician can access the predictive data using any suitable means. Thus, in some embodiments, the present invention provides the further benefit that the clinician, who is not likely to be trained in genetics or molecular biology, need not understand the raw data. The data is presented directly to the clinician in its most useful form. The clinician is then able to immediately utilize the information in order to optimize the care of the subject.

(106) The present invention contemplates any method capable of receiving, processing, and transmitting the information to and from laboratories conducting the assays, information provides, medical personal, and subjects. For example, in some embodiments of the present invention, a sample (e.g., a biopsy or a serum or urine sample) is obtained from a subject and submitted to a profiling service (e.g., clinical lab at a medical facility, genomic profiling business, etc.), located in

any part of the world (e.g., in a country different than the country where the subject resides or where the information is ultimately used) to generate raw data. Where the sample comprises a tissue or other biological sample, the subject may visit a medical center to have the sample obtained and sent to the profiling center, or subjects may collect the sample themselves (e.g., a urine sample) and directly send it to a profiling center. Where the sample comprises previously determined biological information, the information may be directly sent to the profiling service by the subject (e.g., an information card containing the information may be scanned by a computer and the data transmitted to a computer of the profiling center using an electronic communication systems). Once received by the profiling service, the sample is processed and a profile is produced (i.e., expression data), specific for the diagnostic or prognostic information desired for the subject. (107) The profile data is then prepared in a format suitable for interpretation by a treating clinician. For example, rather than providing raw expression data, the prepared format may represent a diagnosis or risk assessment (e.g., expression level of the bladder cancer markers described herein or diagnosis of bladder cancer) for the subject, along with recommendations for particular treatment options. The data may be displayed to the clinician by any suitable method. For example, in some embodiments, the profiling service generates a report that can be printed for the clinician (e.g., at the point of care) or displayed to the clinician on a computer monitor.

(108) In some embodiments, the information is first analyzed at the point of care or at a regional facility. The raw data is then sent to a central processing facility for further analysis and/or to convert the raw data to information useful for a clinician or patient. The central processing facility provides the advantage of privacy (all data is stored in a central facility with uniform security protocols), speed, and uniformity of data analysis. The central processing facility can then control the fate of the data following treatment of the subject. For example, using an electronic communication system, the central facility can provide data to the clinician, the subject, or researchers.

(109) In some embodiments, the subject is able to directly access the data using the electronic communication system. The subject may choose further intervention or counseling based on the results. In some embodiments, the data is used for research use. For example, the data may be used to further optimize the inclusion or elimination of markers as useful indicators of a particular condition or stage of disease or as a companion diagnostic to determine a treatment course of action.

6.2.7. Exemplary Polynucleotides

(110) In some embodiments, polynucleotides are provided. In some embodiments, synthetic polynucleotides are provided. Synthetic polynucleotides, as used herein, refer to polynucleotides that have been synthesized in vitro either chemically or enzymatically. Chemical synthesis of polynucleotides includes, but is not limited to, synthesis using polynucleotide synthesizers, such as OligoPilot (GE Healthcare), ABI 3900 DNA Synthesizer (Applied Biosystems), and the like. Enzymatic synthesis includes, but is not limited, to producing polynucleotides by enzymatic amplification, e.g., PCR. A polynucleotide may comprise one or more nucleotide analogs (i.e., modified nucleotides) discussed herein.

(111) In some embodiments, a polynucleotide is provided that comprises a region that is identical to, or complementary to, at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, at least 15, at least 16, at least 17, at least 18, at least 19, at least 20, at least 21, at least 22, at least 23, at least 24, at least 25, at least 26, at least 27, at least 28, at least 29, or at least 30 contiguous nucleotides of a sequence selected from CRH, IGF2, KRT20, and ANXA10 mRNA. In some embodiments, a polynucleotide is provided that comprises a region that is identical to, or complementary to, a span of 6 to 100, 8 to 100, 8 to 75, 8 to 50, 8 to 40, or 8 to 30 contiguous nucleotides of a sequence selected from CRH, IGF2, KRT20, and ANXA10 mRNA. Nonlimiting exemplary polynucleotides are shown in Tables 1 and 6.

(112) In various embodiments, a polynucleotide comprises fewer than 500, fewer than 300, fewer

than 200, fewer than 150, fewer than 100, fewer than 75, fewer than 50, fewer than 40, or fewer than 30 nucleotides. In various embodiments, a polynucleotide is between 6 and 200, between 8 and 200, between 8 and 150, between 8 and 100, between 8 and 75, between 8 and 50, between 8 and 40, or between 8 and 30 nucleotides long.

(113) In some embodiments, the polynucleotide is a primer. In some embodiments, the primer is labeled with a detectable moiety. In some embodiments, a primer is not labeled. A primer, as used herein, is a polynucleotide that is capable of specifically hybridizing to a target RNA or to a cDNA reverse transcribed from the target RNA or to an amplicon that has been amplified from a target RNA or a cDNA (collectively referred to as “template”), and, in the presence of the template, a polymerase and suitable buffers and reagents, can be extended to form a primer extension product.

(114) In some embodiments, the polynucleotide is a probe. In some embodiments, the probe is labeled with a detectable moiety. A detectable moiety, as used herein, includes both directly detectable moieties, such as fluorescent dyes, and indirectly detectable moieties, such as members of binding pairs. When the detectable moiety is a member of a binding pair, in some embodiments, the probe can be detectable by incubating the probe with a detectable label bound to the second member of the binding pair. In some embodiments, a probe is not labeled, such as when a probe is a capture probe, e.g., on a microarray or bead. In some embodiments, a probe is not extendable, e.g., by a polymerase. In other embodiments, a probe is extendable.

(115) In some embodiments, the polynucleotide is a FRET probe that in some embodiments is labeled at the 5'-end with a fluorescent dye (donor) and at the 3'-end with a quencher (acceptor), a chemical group that absorbs (i.e., suppresses) fluorescence emission from the dye when the groups are in close proximity (i.e., attached to the same probe). In other embodiments, the dye and quencher are not at the ends of the FRET probe. Thus, in some embodiments, the emission spectrum of the dye should overlap considerably with the absorption spectrum of the quencher.

6.2.7.1. Exemplary Polynucleotide Modifications

(116) In some embodiments, the methods of detecting at least one target RNA described herein employ one or more polynucleotides that have been modified, such as polynucleotides comprising one or more affinity-enhancing nucleotide analogs. Modified polynucleotides useful in the methods described herein include primers for reverse transcription, PCR amplification primers, and probes. In some embodiments, the incorporation of affinity-enhancing nucleotides increases the binding affinity and specificity of a polynucleotide for its target nucleic acid as compared to polynucleotides that contain only deoxyribonucleotides, and allows for the use of shorter polynucleotides or for shorter regions of complementarity between the polynucleotide and the target nucleic acid.

(117) In some embodiments, affinity-enhancing nucleotide analogs include nucleotides comprising one or more base modifications, sugar modifications and/or backbone modifications.

(118) In some embodiments, modified bases for use in affinity-enhancing nucleotide analogs include 5-methylcytosine, isocytosine, pseudoisocytosine, 5-bromouracil, 5-propynyluracil, 6-aminopurine, 2-aminopurine, inosine, diaminopurine, 2-chloro-6-aminopurine, xanthine and hypoxanthine.

(119) In some embodiments, affinity-enhancing nucleotide analogs include nucleotides having modified sugars such as 2'-substituted sugars, such as 2'-O-alkyl-ribose sugars, 2'-amino-deoxyribose sugars, 2'-fluoro-deoxyribose sugars, 2'-fluoro-arabinose sugars, and 2'-O-methoxyethyl-ribose (2'MOE) sugars. In some embodiments, modified sugars are arabinose sugars, or d-arabino-hexitol sugars.

(120) In some embodiments, affinity-enhancing nucleotide analogs include backbone modifications such as the use of peptide nucleic acids (PNA; e.g., an oligomer including nucleobases linked together by an amino acid backbone). Other backbone modifications include phosphorothioate linkages, phosphodiester modified nucleic acids, combinations of phosphodiester and phosphorothioate nucleic acid, methylphosphonate, alkylphosphonates, phosphate esters,

alkylphosphonothioates, phosphoramidates, carbamates, carbonates, phosphate triesters, acetamides, carboxymethyl esters, methylphosphorothioate, phosphorodithioate, p-ethoxy, and combinations thereof.

(121) In some embodiments, a polynucleotide includes at least one affinity-enhancing nucleotide analog that has a modified base, at least nucleotide (which may be the same nucleotide) that has a modified sugar, and/or at least one internucleotide linkage that is non-naturally occurring.

(122) In some embodiments, an affinity-enhancing nucleotide analog contains a locked nucleic acid (“LNA”) sugar, which is a bicyclic sugar. In some embodiments, a polynucleotide for use in the methods described herein comprises one or more nucleotides having an LNA sugar. In some embodiments, a polynucleotide contains one or more regions consisting of nucleotides with LNA sugars. In other embodiments, a polynucleotide contains nucleotides with LNA sugars interspersed with deoxyribonucleotides. See, e.g., Frieden, M. et al. (2008) *Curr. Pharm. Des.* 14(11):1138-1142.

6.2.7.2. Exemplary Primers

(123) In some embodiments, a primer is provided. In some embodiments, a primer is identical to, or complementary to, at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, at least 15, at least 16, at least 17, at least 18, at least 19, at least 20, at least 21, at least 22, at least 23, at least 24, at least 25, at least 26, at least 27, at least 28, at least 29, or at least 30 contiguous nucleotides of a sequence selected from CRH, IGF2, KRT20, and ANXA10 mRNA. In some embodiments, a primer is provided that comprises a region that is identical to, or complementary to, a span of 6 to 100, 8 to 100, 8 to 75, 8 to 50, 8 to 40, or 8 to 30 contiguous nucleotides of a sequence selected from CRH, IGF2, KRT20, and ANXA10 mRNA. Nonlimiting exemplary primers are shown in Tables 1 and 6. In some embodiments, a primer may also comprise portions or regions that are not identical or complementary to the target RNA. In some embodiments, a region of a primer that is identical or complementary to a target RNA is contiguous, such that any region of a primer that is not identical or complementary to the target RNA does not disrupt the identical or complementary region.

(124) In some embodiments, a primer comprises a portion that is identically present in a target RNA. In some such embodiments, a primer that comprises a region that is identically present in the target RNA is capable of selectively hybridizing to a cDNA that has been reverse transcribed from the RNA, or to an amplicon that has been produced by amplification of the target RNA or cDNA. In some embodiments, the primer is complementary to a sufficient portion of the cDNA or amplicon such that it selectively hybridizes to the cDNA or amplicon under the conditions of the particular assay being used.

(125) As used herein, “selectively hybridize” means that a polynucleotide, such as a primer or probe, will hybridize to a particular nucleic acid in a sample with at least 5-fold greater affinity than it will hybridize to another nucleic acid present in the same sample that has a different nucleotide sequence in the hybridizing region. Exemplary hybridization conditions are discussed herein, for example, in the context of a reverse transcription reaction or a PCR amplification reaction. In some embodiments, a polynucleotide will hybridize to a particular nucleic acid in a sample with at least 10-fold greater affinity than it will hybridize to another nucleic acid present in the same sample that has a different nucleotide sequence in the hybridizing region.

(126) In some embodiments, a primer is used to reverse transcribe a target RNA, for example, as discussed herein. In some embodiments, a primer is used to amplify a target RNA or a cDNA reverse transcribed therefrom. Such amplification, in some embodiments, is quantitative PCR, for example, as discussed herein. In some embodiments, a primer comprises a detectable moiety.

(127) In some embodiments, primer pairs are provided. Such primer pairs are designed to amplify a portion of a target mRNA, such as CRH, IGF2, KRT20, or ANXA10 mRNA, or an endogenous control RNA, or an exogenous control RNA. In some embodiments, a primer pair is designed to produce an amplicon that is 50 to 1500 nucleotides long, 50 to 1000 nucleotides long, 50 to 750

nucleotides long, 50 to 200 nucleotides long, 50 to 150 nucleotides long, 50 to 100 nucleotides long, 50 to 400 nucleotides long, 50 to 300 nucleotides long, 50 to 200 nucleotides long, 50 to 150 nucleotides long, or 50 to 100 nucleotides long. Nonlimiting exemplary primer pairs are shown in Tables 1 and 6. In some embodiments, a primer pair is designed that spans an intron in the genomic sequence so that the mRNA, without the intron, is more preferably amplified than the genomic sequence. By “spans an intron” is meant that one primer of the primer pair is complementary to a sequence in the mRNA or a cDNA reverse transcribed from the mRNA that is at least partially located 5' to an intron in the genomic sequence and one primer of the primer pair is complementary to a sequence in the mRNA or a cDNA reverse transcribed from the mRNA that is at least partially located 3' to the same intron in the genomic sequence. In some embodiments, one primer of the primer pair is complementary to a sequence in the mRNA or a cDNA reverse transcribed from the mRNA that is located 5' to an intron in the genomic sequence and one primer of the primer pair is complementary to a sequence in the mRNA or a cDNA reverse transcribed from the mRNA that is located 3' to the same intron in the genomic sequence. In some embodiments, one of the primers in the primer pair may be complementary to a sequence in the mRNA or a cDNA reverse transcribed from the mRNA that is spliced together when the intron is removed such that the contiguous complementary sequence is not found in the genomic sequence. A primer pair comprising such a primer is still considered to span an intron.

6.2.7.3. Exemplary Probes

(128) In various embodiments, methods of detecting the presence of bladder cancer comprise hybridizing nucleic acids of a sample with a probe. In some embodiments, the probe comprises a portion that is complementary to a target RNA, such as CRH, IGF2, KRT20 or ANXA10 mRNA. In some embodiments, the probe comprises a portion that is identically present in the target RNA. In some such embodiments, a probe that is complementary to a target RNA is complementary to a sufficient portion of the target RNA such that it selectively hybridizes to the target RNA under the conditions of the particular assay being used. In some embodiments, a probe that is complementary to a target RNA comprises a region that is complementary to at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, at least 15, at least 16, at least 17, at least 18, at least 19, at least 20, at least 21, at least 22, at least 23, at least 24, at least 25, at least 26, at least 27, at least 28, at least 29, or at least 30 contiguous nucleotides of the target RNA, such as CRH, IGF2, KRT20 or ANXA10 mRNA. Nonlimiting exemplary probes are shown in Tables 1 and 6. A probe that is complementary to a target RNA may also comprise portions or regions that are not complementary to the target RNA. In some embodiments, a region of a probe that is complementary to a target RNA is contiguous, such that any region of a probe that is not complementary to the target RNA does not disrupt the complementary region.

(129) In some embodiments, the probe comprises a portion that is identically present in the target RNA, such as CRH, IGF2, KRT20 or ANXA10 mRNA. In some such embodiments, a probe that comprises a region that is identically present in the target RNA is capable of selectively hybridizing to a cDNA that has been reverse transcribed from the RNA, or to an amplicon that has been produced by amplification of the target RNA or cDNA. In some embodiments, the probe is complementary to a sufficient portion of the cDNA or amplicon such that it selectively hybridizes to the cDNA or amplicon under the conditions of the particular assay being used. In some embodiments, a probe that is complementary to a cDNA or amplicon comprises a region that is complementary to at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, at least 15, at least 16, at least 17, at least 18, at least 19, at least 20, at least 21, at least 22, at least 23, at least 24, at least 25, at least 26, at least 27, at least 28, at least 29, or at least 30 contiguous nucleotides of the cDNA or amplicon. A probe that is complementary to a cDNA or amplicon may also comprise portions or regions that are not complementary to the cDNA or amplicon. In some embodiments, a region of a probe that is complementary to a cDNA or amplicon is contiguous, such that any region of a probe that is not complementary to the cDNA or amplicon does not disrupt the complementary region.

(130) In some embodiments, the method of detectably quantifying one or more target RNAs comprises: (a) reverse transcribing a target RNA to produce a cDNA that is complementary to the target RNA; (b) amplifying the cDNA from (a); and (c) detecting the amount of a target RNA using real time RT-PCR and a detection probe (which may be simultaneous with the amplification step (b)).

(131) As described above, in some embodiments, real time RT-PCR detection may be performed using a FRET probe, which includes, but is not limited to, a TaqMan® probe, a Molecular beacon probe and a Scorpion probe. In some embodiments, the real time RT-PCR detection and quantification is performed with a TaqMan® probe, i.e., a linear probe that typically has a fluorescent dye covalently bound at one end of the DNA and a quencher molecule covalently bound at the other end of the DNA. The FRET probe comprises a sequence that is complementary to a region of the cDNA such that, when the FRET probe is hybridized to the cDNA, the dye fluorescence is quenched, and when the probe is digested during amplification of the cDNA, the dye is released from the probe and produces a fluorescence signal. In such embodiments, the amount of target RNA in the sample is proportional to the amount of fluorescence measured during cDNA amplification.

(132) The TaqMan® probe typically comprises a region of contiguous nucleotides having a sequence that is complementary to a region of a target RNA or its complementary cDNA that is reverse transcribed from the target RNA template (i.e., the sequence of the probe region is complementary to or identically present in the target RNA to be detected) such that the probe is specifically hybridizable to the resulting PCR amplicon. In some embodiments, the probe comprises a region of at least 6 contiguous nucleotides having a sequence that is fully complementary to or identically present in a region of a cDNA that has been reverse transcribed from a target RNA template, such as comprising a region of at least 8 contiguous nucleotides, at least 10 contiguous nucleotides, at least 12 contiguous nucleotides, at least 14 contiguous nucleotides, or at least 16 contiguous nucleotides having a sequence that is complementary to or identically present in a region of a cDNA reverse transcribed from a target RNA to be detected.

(133) In some embodiments, the region of the cDNA that has a sequence that is complementary to the TaqMan® probe sequence is at or near the center of the cDNA molecule. In some embodiments, there are independently at least 2 nucleotides, such as at least 3 nucleotides, such as at least 4 nucleotides, such as at least 5 nucleotides of the cDNA at the 5'-end and at the 3'-end of the region of complementarity.

(134) In some embodiments, Molecular Beacons can be used to detect and quantitate PCR products. Like TaqMan® probes, Molecular Beacons use FRET to detect and quantitate a PCR product via a probe having a fluorescent dye and a quencher attached at the ends of the probe. Unlike TaqMan® probes, Molecular Beacons remain intact during the PCR cycles. Molecular Beacon probes form a stem-loop structure when free in solution, thereby allowing the dye and quencher to be in close enough proximity to cause fluorescence quenching. When the Molecular Beacon hybridizes to a target, the stem-loop structure is abolished so that the dye and the quencher become separated in space and the dye fluoresces. Molecular Beacons are available, e.g., from Gene Link™ (see genelink.com/newsite/products/mbintro.asp).

(135) In some embodiments, Scorpion probes can be used as both sequence-specific primers and for PCR product detection and quantitation. Like Molecular Beacons, Scorpion probes form a stem-loop structure when not hybridized to a target nucleic acid. However, unlike Molecular Beacons, a Scorpion probe achieves both sequence-specific priming and PCR product detection. A fluorescent dye molecule is attached to the 5'-end of the Scorpion probe, and a quencher is attached to the 3'-end. The 3' portion of the probe is complementary to the extension product of the PCR primer, and this complementary portion is linked to the 5'-end of the probe by a non-amplifiable moiety. After the Scorpion primer is extended, the target-specific sequence of the probe binds to its complement within the extended amplicon, thus opening up the stem-loop structure and allowing

the dye on the 5'-end to fluoresce and generate a signal. Scorpion probes are available from, e.g., Premier Biosoft International (see premierbiosoft.com/tech_notes/Scorpion.html).

(136) In some embodiments, labels that can be used on the FRET probes include colorimetric and fluorescent dyes such as Alexa Fluor dyes, BODIPY dyes, such as BODIPY FL; Cascade Blue; Cascade Yellow; coumarin and its derivatives, such as 7-amino-4-methylcoumarin, aminocoumarin and hydroxycoumarin; cyanine dyes, such as Cy3 and Cy5; eosins and erythrosins; fluorescein and its derivatives, such as fluorescein isothiocyanate; macrocyclic chelates of lanthanide ions, such as Quantum Dye™; Marina Blue; Oregon Green; rhodamine dyes, such as rhodamine red, tetramethylrhodamine and rhodamine 6G; Texas Red; fluorescent energy transfer dyes, such as thiazole orange-ethidium heterodimer; and, TOTAB.

(137) Specific examples of dyes include, but are not limited to, those identified above and the following: Alexa Fluor 350, Alexa Fluor 405, Alexa Fluor 430, Alexa Fluor 488, Alexa Fluor 500, Alexa Fluor 514, Alexa Fluor 532, Alexa Fluor 546, Alexa Fluor 555, Alexa Fluor 568, Alexa Fluor 594, Alexa Fluor 610, Alexa Fluor 633, Alexa Fluor 647, Alexa Fluor 660, Alexa Fluor 680, Alexa Fluor 700, and, Alexa Fluor 750; amine-reactive BODIPY dyes, such as BODIPY 493/503, BODIPY 530/550, BODIPY 558/568, BODIPY 564/570, BODIPY 576/589, BODIPY 581/591, BODIPY 630/650, BODIPY 650/655, BODIPY FL, BODIPY R6G, BODIPY TMR, and, BODIPY-TR; Cy3, Cy5, 6-FAM, Fluorescein Isothiocyanate, HEX, 6-JOE, Oregon Green 488, Oregon Green 500, Oregon Green 514, Pacific Blue, REG, Rhodamine Green, Rhodamine Red, Renographin, ROX, SYPRO, TAMRA, 2',4',5',7'-Tetrabromosulfonefluorescein, and TET.

(138) Examples of dye/quencher pairs (i.e., donor/acceptor pairs) include, but are not limited to, fluorescein/tetramethylrhodamine; IAEDANS/fluorescein; EDANS/dabcyl; fluorescein/fluorescein; BODIPY FL/BODIPY FL; fluorescein/QSY 7 or QSY 9 dyes. When the donor and acceptor are the same, FRET may be detected, in some embodiments, by fluorescence depolarization. Certain specific examples of dye/quencher pairs (i.e., donor/acceptor pairs) include, but are not limited to, Alexa Fluor 350/Alexa Fluor 488; Alexa Fluor 488/Alexa Fluor 546; Alexa Fluor 488/Alexa Fluor 555; Alexa Fluor 488/Alexa Fluor 568; Alexa Fluor 488/Alexa Fluor 594; Alexa Fluor 488/Alexa Fluor 647; Alexa Fluor 546/Alexa Fluor 568; Alexa Fluor 546/Alexa Fluor 594; Alexa Fluor 546/Alexa Fluor 647; Alexa Fluor 555/Alexa Fluor 594; Alexa Fluor 555/Alexa Fluor 647; Alexa Fluor 568/Alexa Fluor 647; Alexa Fluor 594/Alexa Fluor 647; Alexa Fluor 350/QSY 35; Alexa Fluor 350/dabcyl; Alexa Fluor 488/QSY 35; Alexa Fluor 488/dabcyl; Alexa Fluor 488/QSY 7 or QSY 9; Alexa Fluor 555/QSY 7 or QSY 9; Alexa Fluor 568/QSY 7 or QSY 9; Alexa Fluor 568/QSY 21; Alexa Fluor 594/QSY 21; and Alexa Fluor 647/QSY 21. In some instances, the same quencher may be used for multiple dyes, for example, a broad spectrum quencher, such as an Iowa Black® quencher (Integrated DNA Technologies, Coralville, IA) or a Black Hole Quencher™ (BHQ™; Sigma-Aldrich, St. Louis, MO).

(139) In some embodiments, for example, in a multiplex reaction in which two or more moieties (such as amplicons) are detected simultaneously, each probe comprises a detectably different dye such that the dyes may be distinguished when detected simultaneously in the same reaction. One skilled in the art can select a set of detectably different dyes for use in a multiplex reaction.

(140) Specific examples of fluorescently labeled ribonucleotides useful in the preparation of RT-PCR probes for use in some embodiments of the methods described herein are available from Molecular Probes (Invitrogen), and these include, Alexa Fluor 488-5-UTP, Fluorescein-12-UTP, BODIPY FL-14-UTP, BODIPY TMR-14-UTP, Tetramethylrhodamine-6-UTP, Alexa Fluor 546-14-UTP, Texas Red-5-UTP, and BODIPY TR-14-UTP. Other fluorescent ribonucleotides are available from Amersham Biosciences (GE Healthcare), such as Cy3-UTP and Cy5-UTP.

(141) Examples of fluorescently labeled deoxyribonucleotides useful in the preparation of RT-PCR probes for use in the methods described herein include Dinitrophenyl (DNP)-1'-dUTP, Cascade Blue-7-dUTP, Alexa Fluor 488-5-dUTP, Fluorescein-12-dUTP, Oregon Green 488-5-dUTP, BODIPY FL-14-dUTP, Rhodamine Green-5-dUTP, Alexa Fluor 532-5-dUTP, BODIPY TMR-14-

dUTP, Tetramethylrhodamine-6-dUTP, Alexa Fluor 546-14-dUTP, Alexa Fluor 568-5-dUTP, Texas Red-12-dUTP, Texas Red-5-dUTP, BODIPY TR-14-dUTP, Alexa Fluor 594-5-dUTP, BODIPY 630/650-14-dUTP, BODIPY 650/665-14-dUTP; Alexa Fluor 488-7-OBEA-dCTP, Alexa Fluor 546-16-OBEA-dCTP, Alexa Fluor 594-7-OBEA-dCTP, Alexa Fluor 647-12-OBEA-dCTP. Fluorescently labeled nucleotides are commercially available and can be purchased from, e.g., Invitrogen.

(142) In some embodiments, dyes and other moieties, such as quenchers, are introduced into polynucleotide used in the methods described herein, such as FRET probes, via modified nucleotides. A “modified nucleotide” refers to a nucleotide that has been chemically modified, but still functions as a nucleotide. In some embodiments, the modified nucleotide has a chemical moiety, such as a dye or quencher, covalently attached, and can be introduced into a polynucleotide, for example, by way of solid phase synthesis of the polynucleotide. In other embodiments, the modified nucleotide includes one or more reactive groups that can react with a dye or quencher before, during, or after incorporation of the modified nucleotide into the nucleic acid. In specific embodiments, the modified nucleotide is an amine-modified nucleotide, i.e., a nucleotide that has been modified to have a reactive amine group. In some embodiments, the modified nucleotide comprises a modified base moiety, such as uridine, adenosine, guanosine, and/or cytosine. In specific embodiments, the amine-modified nucleotide is selected from 5-(3-aminoallyl)-UTP; 8-[(4-amino)butyl]-amino-ATP and 8-[(6-amino)butyl]-amino-ATP; N6-(4-amino)butyl-ATP, N6-(6-amino)butyl-ATP, N4-[2,2-oxy-bis-(ethylamine)]-CTP; N6-(6-Amino)hexyl-ATP; 8-[(6-Amino)hexyl]-amino-ATP; 5-propargylamino-CTP, 5-propargylamino-UTP. In some embodiments, nucleotides with different nucleobase moieties are similarly modified, for example, 5-(3-aminoallyl)-GTP instead of 5-(3-aminoallyl)-UTP. Many amine modified nucleotides are commercially available from, e.g., Applied Biosystems, Sigma, Jena Bioscience and TriLink.

(143) Exemplary detectable moieties also include, but are not limited to, members of binding pairs. In some such embodiments, a first member of a binding pair is linked to a polynucleotide. The second member of the binding pair is linked to a detectable label, such as a fluorescent label. When the polynucleotide linked to the first member of the binding pair is incubated with the second member of the binding pair linked to the detectable label, the first and second members of the binding pair associate and the polynucleotide can be detected. Exemplary binding pairs include, but are not limited to, biotin and streptavidin, antibodies and antigens, etc.

(144) In some embodiments, multiple target RNAs are detected in a single multiplex reaction. In some such embodiments, each probe that is targeted to a unique cDNA is spectrally distinguishable when released from the probe. Thus, each target RNA is detected by a unique fluorescence signal.

(145) One skilled in the art can select a suitable detection method for a selected assay, e.g., a real-time RT-PCR assay. The selected detection method need not be a method described above, and may be any method.

6.3. Exemplary Compositions and Kits

(146) In another aspect, compositions are provided. In some embodiments, compositions are provided for use in the methods described herein.

(147) In some embodiments, compositions are provided that comprise at least one target RNA-specific primer. The term “target RNA-specific primer” encompasses primers that have a region of contiguous nucleotides having a sequence that is (i) identically present in a target RNA, such as CRH, IGF2, KRT20, or ANXA10 mRNA, or (ii) complementary to the sequence of a region of contiguous nucleotides found in a target RNA, such as CRH, IGF2, KRT20, or ANXA10 mRNA. In some embodiments, a composition is provided that comprises at least one pair of target RNA-specific primers. The term “pair of target RNA-specific primers” encompasses pairs of primers that are suitable for amplifying a defined region of a target RNA, such as CRH, IGF2, KRT20, or ANXA10 mRNA. A pair of target RNA-specific primers typically comprises a first primer that

comprises a sequence that is identical to the sequence of a region of a target RNA (although the primer will typically comprise DNA or modified nucleosides rather than RNA) and a second primer that comprises a sequence that is complementary to a region of a target RNA. A pair of primers is typically suitable for amplifying a region of a target mRNA that is 50 to 1500 nucleotides long, 50 to 1000 nucleotides long, 50 to 750 nucleotides long, 50 to 500 nucleotides long, 50 to 400 nucleotides long, 50 to 300 nucleotides long, 50 to 200 nucleotides long, 50 to 150 nucleotides long, or 50 to 100 nucleotides long. Nonlimiting exemplary primers, and pairs of primers, are shown in Tables 1 and 6.

(148) In some embodiments, a composition comprises four pairs of target RNA-specific primers, one pair for amplifying each of CRH, IGF2, KRT20, and ANXA10 mRNA. In some embodiments, a composition additionally comprises a pair of target RNA-specific primers for amplifying an endogenous control RNA and/or one pair of target RNA-specific primers for amplifying an exogenous control RNA.

(149) In some embodiments, a composition comprises at least one target RNA-specific probe. The term “target RNA-specific probe” encompasses probes that have a region of contiguous nucleotides having a sequence that is (i) identically present in a target RNA, such as CRH, IGF2, KRT20, or ANXA10 mRNA, or (ii) complementary to the sequence of a region of contiguous nucleotides found in a target RNA, such as CRH, IGF2, KRT20, or ANXA10 mRNA. Nonlimiting exemplary target-specific probes are shown in Tables 1 and 6.

(150) In some embodiments, a composition (including a composition described above that comprises four or more pairs of target RNA-specific primers) comprises four probes, one probe for detecting each of CRH, IGF2, KRT20, and ANXA10 mRNA. In some embodiments, a composition additionally comprises a probe for detecting an endogenous control RNA and/or a probe for detecting an exogenous control RNA.

(151) In some embodiments, a composition is an aqueous composition. In some embodiments, the aqueous composition comprises a buffering component, such as phosphate, tris, HEPES, etc., and/or additional components, as discussed below. In some embodiments, a composition is dry, for example, lyophilized, and suitable for reconstitution by addition of fluid. A dry composition may include one or more buffering components and/or additional components.

(152) In some embodiments, a composition further comprises one or more additional components. Additional components include, but are not limited to, salts, such as NaCl, KCl, and MgCl₂; polymerases, including thermostable polymerases such as Taq; dNTPs; reverse transcriptases, such as MMLV reverse transcriptase; RNase inhibitors; bovine serum albumin (BSA) and the like; reducing agents, such as β -mercaptoethanol; EDTA and the like; etc. One skilled in the art can select suitable composition components depending on the intended use of the composition.

(153) In some embodiments, compositions are provided that comprise at least one polynucleotide for detecting at least one target RNA. In some embodiments, the polynucleotide is used as a primer for a reverse transcriptase reaction. In some embodiments, the polynucleotide is used as a primer for amplification. In some embodiments, the polynucleotide is used as a primer for RT-PCR. In some embodiments, the polynucleotide is used as a probe for detecting at least one target RNA. In some embodiments, the polynucleotide is detectably labeled. In some embodiments, the polynucleotide is a FRET probe. In some embodiments, the polynucleotide is a TaqMan® probe, a Molecular Beacon, or a Scorpion probe.

(154) In some embodiments, a composition comprises at least one FRET probe having a sequence that is identically present in, or complementary to a region of, CRH, IGF2, KRT20, or ANXA10 mRNA. In some embodiments, a FRET probe is labeled with a donor/acceptor pair such that when the probe is digested during the PCR reaction, it produces a unique fluorescence emission that is associated with a specific target RNA. In some embodiments, when a composition comprises multiple FRET probes, each probe is labeled with a different donor/acceptor pair such that when the probe is digested during the PCR reaction, each one produces a unique fluorescence emission

that is associated with a specific probe sequence and/or target RNA. In some embodiments, the sequence of the FRET probe is complementary to a target region of a target RNA. In other embodiments, the FRET probe has a sequence that comprises one or more base mismatches when compared to the sequence of the best-aligned target region of a target RNA.

(155) In some embodiments, a composition comprises a FRET probe consisting of at least 8, at least 9, at least 10, at least 11, at least 13, at least 14, at least 15, at least 16, at least 17, at least 18, at least 19, at least 20, at least 21, at least 22, at least 23, at least 24, or at least 25 nucleotides, wherein at least a portion of the sequence is identically present in, or complementary to a region of, CRH, IGF2, KRT20, or ANXA10 mRNA. In some embodiments, at least 8, at least 9, at least 10, at least 11, at least 13, at least 14, at least 15, at least 16, at least 17, at least 18, at least 19, at least 20, at least 21, at least 22, at least 23, at least 24, or at least 25 nucleotides of the FRET probe are identically present in, or complementary to a region of, CRH, IGF2, KRT20, or ANXA10 mRNA. In some embodiments, the FRET probe has a sequence with one, two or three base mismatches when compared to the sequence or complement of small CRH, IGF2, KRT20, or ANXA10 mRNA.

(156) In some embodiments, a kit comprises a polynucleotide discussed above. In some embodiments, a kit comprises at least one primer and/or probe discussed above. In some embodiments, a kit comprises at least one polymerase, such as a thermostable polymerase. In some embodiments, a kit comprises dNTPs. In some embodiments, kits for use in the real time RT-PCR methods described herein comprise one or more target RNA-specific FRET probes and/or one or more primers for reverse transcription of target RNAs and/or one or more primers for amplification of target RNAs or cDNAs reverse transcribed therefrom.

(157) In some embodiments, one or more of the primers and/or probes is “linear”. A “linear” primer refers to a polynucleotide that is a single stranded molecule, and typically does not comprise a short region of, for example, at least 3, 4 or 5 contiguous nucleotides, which are complementary to another region within the same polynucleotide such that the primer forms an internal duplex. In some embodiments, the primers for use in reverse transcription comprise a region of at least 4, such as at least 5, such as at least 6, such as at least 7 or more contiguous nucleotides at the 3'-end that has a sequence that is complementary to region of at least 4, such as at least 5, such as at least 6, such as at least 7 or more contiguous nucleotides at the 5'-end of a target RNA.

(158) In some embodiments, a kit comprises one or more pairs of linear primers (a “forward primer” and a “reverse primer”) for amplification of a cDNA reverse transcribed from a target RNA, such as CRH, IGF2, KRT20, or ANXA10 mRNA. Accordingly, in some embodiments, a first primer comprises a region of at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, at least 15, at least 16, at least 17, at least 18, at least 19, at least 20, at least 21, at least 22, at least 23, at least 24, or at least 25 contiguous nucleotides having a sequence that is identical to the sequence of a region of at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, at least 15, at least 16, at least 17, at least 18, at least 19, at least 20, at least 21, at least 22, at least 23, at least 24, or at least 25 contiguous nucleotides at a first location in the mRNA.

Furthermore, in some embodiments, a second primer comprises a region of at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, at least 15, at least 16, at least 17, at least 18, at least 19, at least 20, at least 21, at least 22, at least 23, at least 24, or at least 25 contiguous nucleotides having a sequence that is complementary to the sequence of a region of at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, at least 15, at least 16, at least 17, at least 18, at least 19, at least 20, at least 21, at least 22, at least 23, at least 24, or at least 25 contiguous nucleotides at a second location in the mRNA, such that a PCR reaction using the two primers results in an amplicon extending from the first location of the mRNA to the second location of the mRNA.

(159) In some embodiments, the kit comprises at least two, at least three, or at least four sets of primers, each of which is for amplification of a cDNA that is reverse transcribed from a different target RNA, including CRH, IGF2, KRT20, and ANXA10 mRNA. In some embodiments, the kit

further comprises at least one set of primers for amplifying a control RNA, such as an endogenous control and/or an exogenous control.

(160) In some embodiments, probes and/or primers for use in the compositions described herein comprise deoxyribonucleotides. In some embodiments, probes and/or primers for use in the compositions described herein comprise deoxyribonucleotides and one or more nucleotide analogs, such as LNA analogs or other duplex-stabilizing nucleotide analogs described above. In some embodiments, probes and/or primers for use in the compositions described herein comprise all nucleotide analogs. In some embodiments, the probes and/or primers comprise one or more duplex-stabilizing nucleotide analogs, such as LNA analogs, in the region of complementarity.

(161) In some embodiments, the kits for use in real time RT-PCR methods described herein further comprise reagents for use in the reverse transcription and amplification reactions. In some embodiments, the kits comprise enzymes such as reverse transcriptase, and a heat stable DNA polymerase, such as Taq polymerase. In some embodiments, the kits further comprise deoxyribonucleotide triphosphates (dNTP) for use in reverse transcription and amplification. In further embodiments, the kits comprise buffers optimized for specific hybridization of the probes and primers.

(162) The following examples are for illustration purposes only, and are not meant to be limiting in any way.

7. EXAMPLES

7.1. Example 1: Detection of High Grade and Low Grade Bladder Cancer

(163) More than 30 mRNA markers and 20 microRNA markers were evaluated in both bladder tissue and urine samples to determine the most accurate panel for detection of bladder cancer. Based on results from over 200 urine samples using eight markers and the GeneXpert system (Cepheid, Sunnyvale, CA), a panel consisting of CRH, IGF2, KRT20, and ANXA10 mRNA markers (with or without at least one endogenous and/or at least one exogenous control) was selected.

(164) Various reaction compositions were designed for use in the GeneXpert® system, for detecting various combinations of CRH, IGF2, KRT20, and ANXA10 mRNA. Table 1 shows the sequences of the primers and probes used to detect each of the target RNAs by quantitative RT-PCR in the various reaction compositions.

(165) TABLE-US-00002

TABLE	1	Primer and probe sequences	SEQ	Reagent	Formul.
oligo	name	target	sequence	ID	NO ("TSR")
					ABLa3a4 ABL
					GATCAACACTGCTTCTGATGGCAA
					8 CL3, CL4, CL1
					PrmrFwd4 ABLa3a4 ABL
					CCACCGTTGAATGATGATGAACCAA
					9 CL3, CL4, CL1
					PrmrRev1 ABLa3a4 Probe
					1 ABL F4-CCTCCGAGAGCCGCTTCAAC-Q4
					10 CL3 ABL probe F6 ABL F6-
					CCTCCGAGAGCCGCTTCAAC-Q6
					11 CL4 ABL probe F1 ABL F1-
					CCTCCGAGAGCCGC(T-dabsyl)TCAAC-Q1
					12 CL1 KRT20 For KRT20
					TTGAAGAGCTGCGAAGTCAGAT
					13 CL3, CL4 KRT20 Rev KRT20
					TGAAGTCCTCAGCAGCCAGTT
					14 CL3, CL4 KRT20 Probe (F3) KRT20 F3-
					TCAACTGCAAAATGCTCGGTGTGTCC-Q3
					15 CL3, CL4 IGF2 For_4 IGF2
					CGCGGCTTCTACTTCAGCAG
					16 CL3, CL4 IGF2 Rev_4 IGF2
					GCGGAAACAGCACTCCTCAA
					17 CL3, CL4 IGF2 Probe_2 IGF2 F5-
					TGTGAGCCGTTCGAGCCGTG-Q5
					18 CL3, CL4 CRH_For CRH
					ACCCGGCTCACCTGCGAA
					19 CL3, CL4, CL1 CRH_Rev CRH
					GGACTCCCGCGGACACAA
					20 CL3, CL4, CL1 CRH_probe
					3 CRH F2-TCCTGGGAAGCGAGTGCCCCTAA-Q2
					21 CL3, CL1 CRH_probe_F1 CRH F1-
					CCTGGGAAGCGAG(T-Dabsyl)GCCCCTAA-Q1
					22 CL4 Armored RNA ® exogenous
					GGCTATTCTCCTCTTGGCAGAT
					23 CL1 Fwd control Armored RNA ® exogenous
					TGCTTGAGCTCCAGTCCCTAAG
					24 CL1 Rev control Armored exogenous F6-
					AGCCGAGAAGGCGGAGTCTGGC-Q6
					25 CL1 RNA ®_Probe control ANXA10-FW

ANXA10 GTGAAACAAGTTTATGCAATCGATCAA 26 CL1 ANXA10-RV3 ANXA10

GATTGAAATTGGGAGCTGGGAA 27 CL1 ANXA10-F3 ANXA10 F3-

TCATCCCTGAGGTAAACAATTACCATCAA-Q3 28 CL1

F1 through F6 are detectably different dyes that can be detected and distinguished simultaneously in a multiplex reaction, and Q1 to Q6 are quenchers (in the present example, Q2, Q4, Q5, and Q6 are the same quencher).

(166) The final primer and probe compositions of three different reaction compositions are shown in Table 2.

(167) TABLE-US-00003 TABLE 2 Primers and probes in TSR CL3, CL4, and CL1 Final conc.

Final conc.	Final conc.	Target	Label	Purpose	forw. primer	rev primer	probe
TSR CL3	ABL	F4	Normalization	400 nM	400 nM	150 nM (endogenous control)	KRT20 F3 Bladder cancer marker
400 nM	400 nM	75 nM	IGF2 F5	Bladder cancer marker	400 nM	400 nM	200 nM CRH F2
Bladder cancer marker	400 nM	400 nM	200 nM	TSR CL4	ABL	F6	Normalization
400 nM	400 nM	400 nM	400 nM	400 nM (endogenous control)	KRT20	F3	Bladder cancer marker
400 nM	400 nM	75 nM	IGF2 F5	Bladder cancer marker	400 nM	400 nM	300 nM CRH F1
Bladder cancer marker	400 nM	400 nM	300 nM	CRH F1	Bladder cancer marker	400 nM	400 nM
600 nM	TSR CL1	ABL	F1	Normalization	400 nM	400 nM	600 nM (endogenous control)
Armored F6	Exogenous control	400 nM	400 nM	400 nM	RNA ®	ANXA10	F3
Bladder cancer marker	400 nM	400 nM	75 nM	CRH F2	Bladder cancer marker	400 nM	400 nM
300 nM							

(168) Each reaction contained 50-90 mM KCl, 3-5 mM MgCl₂, 400-825 µM dNTPs, 20 mM Tris, pH 8.5, 0.01% sodium azide, and 0.9 units/µl of RNase inhibitor. MMLV reverse transcriptase (0.375 units/µl) and AptaTaq (0.25 units/µl; Roche) were used for reverse transcription and amplification, respectively. TSR CL1 included an Armored RNA® exogenous control (SEQ ID NO: 47; Asuragen, Austin, TX).

(169) For each sample to be tested, 5 mL of voided urine was added to 5 mL preservative (3.5M guanidine HCl, 1% N-acetyl-L-cysteine, 25 mM sodium citrate, and 2.5% Tween-20, pH 3.2), preferably within 1 hour of sample collection. The preserved samples were transported on ice and stored at 4° C. Clinical information for each sample was provided by the collection sites. The number of red blood cells per millilitre was determined by microscopic evaluation.

(170) Prior to use, the preserved urine was inverted three times to mix. 1.2 mL of preserved urine was loaded into a GeneXpert cartridge for analysis. The cartridge contained a 0.8 µm filter to capture urothelial cells. The captured cells were washed and lysed using sonication (2-15 seconds, 8-16 µm at 36 kHz) within the cartridge. The lysate was then used to reconstitute the reagents used for real-time RT-PCR (described above). The reaction cycle used was: 10 minutes at 45° C., followed by 2 minutes at 95° C., and then 45 cycles of (a) 5 seconds at 95° C., 20 seconds at 60° C., and 20 seconds at 72° C., using a GeneXpert® cartridge in a GeneXpert® system. Delta Ct (ΔCt) was calculated as Ct (ABL)–Ct (marker). The ΔCt cutoff was set as the ΔCt that gave at least 95% specificity with samples from patients not expected to have bladder cancer (data not shown). A ΔCt above the ΔCt cutoff for any one of the markers was considered a positive result, indicative of the presence of bladder cancer.

(171) Some samples were also tested using UroVysion® (Abbott Laboratories, Abbott Park, IL). The results of that experiment are shown in Table 3 (high grade bladder cancer) and Table 4 (low grade bladder cancer). ΔCts above the threshold, indicating a positive result, are highlighted. Each of the three TSR lots, CL3, CL4, and CL1, detected 100% of high grade bladder cancer samples, as did UroVysion.

(172) For low grade bladder cancer, the detection rate was 37% (7/19), compared to only 16% (3/19) for UroVysion®.

(173) TABLE-US-00004 TABLE 3 Detection of high grade bladder cancer history CIC TSR of

TSR lot	CL3	TSR lot	CL4	lot	CL1	Sample	UroVysion ®	bladder	CRH	KRT20	IGF2	KRT20	IGF2	CRH	ANXA10	CRH	GX	ID	stage	grade	Result	cytology	cancer			
-10	2.2	-1	4	0.5	-5	-0.5	-3	result	67001	pTa	high	positive	suspicious	yes	-20	4.9	-20	4.8	1.1	-20	-1.6	-3.6	positive	67006	pT1	high

positive positive no -0.7 4.4 -6.2 5.7 -2.8 -0.6 -20 2.2 positive 67009 pT2 high positive positive no -20 4.9 -0.4 4.9 1.5 -20 1.1 -20 positive 75211 pT2 high positive negative no -0.8 1.4 -4.1 positive 75216 pTa, high positive suspicious yes -3.5 2 -0.8 0.9 -0.6 -3.4 positive CIC 75218 pTa high positive negative no 0.9 3.8 -0.9 3.8 -1.6 0.5 positive 75245 pTa high positive NA no 20 1.6 6.7 0.4 6.2 20 1.9 8.2 positive 75247 CIS high positive atypical no -1.4 3.4 2.7 3.6 3.7 -3.8 -0.2 0.4 positive 75248 pT1/ high positive atypical no -20 4.3 0 4.8 1 -20 0.4 -5.1 positive CIS 75249 pT1 high positive negative no -20 3.3 2.1 3.4 2.5 -20 3.7 -20 positive 75258 CIS, high positive atypical no 4.2 -3.7 0.6 -2 1.9 positive pTa 75246 positive positive no -20 4.3 5.3 4.4 5.9 9.1 -1 3.4 positive

(174) TABLE-US-00005 TABLE 4 Detection of low grade bladder cancer history CIC TSR of TSR lot CL3 TSR lot CL4 lot CL1 Sample UroVysion® bladder CRH KRT20 IGF2 KRT20 IGF2 CRH ANXA10 CRH GX ID stage grade Result cytology cancer -10 2.2 -1 4 0.5 -5 -0.5 -3 result 67002 pTa low negative atypical yes -20 3.9 -3.8 3.3 -1.5 -20 -1.7 -20 positive 67003 pTa low negative negative no -20 -20 -3.2 -20 -20 -20 -20 -20 negative 67004 pTa low negative negative no -2 4.7 -2.3 5.2 0.1 -1.7 2.4 2.2 positive 67010 pTa low positive atypical no -20 -5.3 -8.8 -3.8 -6.7 -20 -5 -20 negative 67011 pTa low positive atypical no -1.3 4.6 1.9 4 2.4 -2 -1.3 1 positive 67018 pTa low negative atypical yes -20 -20 -4.4 -20 -20 -20 -20 -20 negative 67050 pTa low negative atypical yes -20 -0.4 3.4 1.9 3.3 -20 -0.5 -20 positive 67100 pTa low borderline atypical yes -2.9 -2.7 -20 -20 -20 negative 75161 pTa low negative atypical yes -20 -3 -4.7 -3.6 -1.3 -20 -5.7 -20 negative 75183 pTa low positive negative yes -1.5 3.1 -2 3.4 -0.5 0.7 -4.1 2.3 positive 75184 pTa low inconclusive negative yes -20 -0.6 1.3 -1.5 1.8 -20 -20 -20 positive 75185 pTa low negative negative no -20 3 5 1.3 4.7 -20 1.2 -4.2 positive 75191 pTa low negative negative yes -20 -0.4 -3.4 1.1 -0.5 -20 -4.1 -20 negative 75202 pTa low negative negative yes -20 0.3 -2.3 -0.4 -1.5 -20 -2.4 -20 negative 75236 pTa low negative negative no -20 -5.7 -7.9 -20 -8.9 -20 -3.2 -20 negative 75251 pTa low negative negative yes -20 -3.6 -5.1 -1 -4.4 -20 -20 -20 negative 75257 pTa low negative negative no -2.1 -0.9 -20 -20 -20 negative 75265 pTa low negative negative yes 1.4 -20 -20 -20 -20 negative

(175) A summary of the sensitivity for high grade bladder cancer and low grade bladder cancer, and the specificity in patients with a low risk of bladder cancer, is shown in Table 5 for each of the individual markers tested in Tables 3 and 4.

(176) TABLE-US-00006 TABLE 5 Summary of sensitivity and specificity of individual markers Sensitivity, Sensitivity, Specificity, high grade low grade low risk of Marker bladder cancer bladder cancer bladder cancer CRH 15/30 (50%) 9/53 (17%) 220/221 (99%) KRT20 11/21 (52%) 6/34 (18%) 144/145 (99%) IGF2 13/21 (62%) 8/34 (24%) 144/145 (99%) ANXA10 5/9 (56%) 2/19 (11%) 74/76 (97%) 4 marker combo 12/12 (100%) 7/19 (37%) 83/88 (94%)

(177) Exemplary alternative primers and probes for detecting the four markers, KRT20, IGF2, CRH, and ANXA10, are shown in Table 6. Table 6 also shows an exemplary set of primers and probes for detecting an exogenous control and an endogenous control, ABL. The dyes and quenchers shown in Table 6 are generic, and two or more of quenchers Q1 to Q6 may be the same. One skilled in the art could select a suitable set, for example, a set of detectably different dyes for use in a multiplex assay. The predicted amplicon length for each set of primers is also shown, as well as the length of any intervening intron(s) between the primer sites on the genomic copy of the target.

(178) TABLE-US-00007 TABLE 6 Primer and probe sequences amplicon intron SEQ length length name 5' mod sequence 3' mod ID NO (bp) (bp) KRT20 For_3 CGACTACAGTGCATATTACAGACAA 29 113 2142 KRT20 Rev_2 CAGCAGCCAGTTTAGCATTATCAA 30 KRT20 Probe F1 TCAACTGCAAAA(T-dabsyl)GCT Q1 31 CGGTGTGTCC IGF2 For_5 GGACCGCGGCTTCTACTTCA 32 95 1701 IGF2 Rev_5 CCAGGTCACAGCTGCGGAA 33 IGF2 Probe_2_F4 F4 TGTGAGCCGTCGCAGCCGTG Q4 34 CRH_For 4 TGCGAAGCGCCTGGGAAGC 35 66

801 CRH_Rev GGACTCCCAGGACACAA 36 CRH_probe_F2 F2
 TGCCCCTAACATGCGGCTGCC Q2 37 ANXA10_For_3 TCAGCGCTGCAATGCACAA 38 122
 22, 947 ANXA10_For_4 CTGCAATGCACAAAGGATGA 48 117 22, 947 ANXA10_Rev_5
 GGCCAGCCATCACATCTTTGAA 39 ANXA10_Probe_3 F3
 TAGAGCATGTATGGCCGGGACCT Q3 40 ABLa3a4 GATCAACACTGCTTCTGATGGCAA 41
 92 7666 PrmrFwd4 ABLa3a4 CCACCGTTGAATGATGATGAACCAA 42 PrmrRev1 ABL
 Probe F5 F5 CCTCCGAGAGCCGCTTCAAC Q5 43 Armored RNA ®
 GGCTATTCTCCTCTTGGCAGAT 44 101 NA Fwd Armored RNA ®
 TGCTTGAGCTCCAGTCCCTAAG 45 Rev Armored RNA ® F6
 AGCCGAGAAGGCGGAGTCTGGC Q6 46 Probe

7.2. Example 2: Assay Sensitivity for Detecting Bladder Cancer Using Marker Panel KRT20, IGF2, CRH, and ANXA10 in a Larger Cohort

(179) Urine samples were collected from subjects at seven different sites. Eligibility criteria for inclusion in the study included: 18 years or older; Documented informed consent as required by the reviewing IRB or HREC, and a signed Experimental Subjects Bill of Rights for patients in California; At least one of the following criteria: A history or recurrence of bladder cancer; A referral for cystoscopy evaluation due to micro- or gross-hematuria in urine; A referral for urology evaluation, but no previous history of bladder cancer or clinical evidence of bladder cancer; Consent to provide at least 15 ml voided urine in addition to that required for standard of care; Consent to allow pathology results for any biopsy specimens taken during cystoscopy procedure and other medical records to be reported.

Exclusion criteria included only under 18 years of age and first voided urine. Patients currently or previously treated with Bacillus Calmette-Guerin (BCG) and patients currently or previously treated with intravesical therapy or transurethral resection of bladder or radiation therapy for bladder cancer were eligible for the study. In addition, repeat enrollment during the course of the study was also permitted.

(180) Two of the collection sites provided the results of UroVysion® analysis on the urine samples. For each sample to be tested 15 mL of voided urine was added to 15 mL of preservative (3.5M guanidine HCl, 1% N-acetyl-L-cysteine, 25 mM sodium citrate, and 2.5% Tween-20, pH 3.2), preferably within 1 hour of sample collection. The preserved samples were transported on ice and stored at 4° C. Clinical information for each sample was provided by the collection sites.

(181) Prior to use, the preserved urine was inverted three times to mix. 4 mL of preserved urine was loaded into a GeneXpert cartridge for analysis. The cartridge contained a 0.8 µm filter to capture urothelial cells. The captured cells were washed and lysed using sonication (2-15 seconds, 8-16 µm at 36 kHz) within the cartridge. The lysate was then used to reconstitute the reagents used for real-time RT-PCR (described above). The reaction cycle used was: 10 minutes at 45° ° C., followed by 2 minutes at 95° C., and then 45 cycles of (a) 5 seconds at 95° C., 20 seconds at 60° C., and 20 seconds at 72° C., using a GeneXpert® cartridge in a GeneXpert® system. For ANXA10, KRT20 and IGF2, delta Ct (Δ Ct) was calculated as Ct (ABL)–Ct (marker). The Δ Ct cutoff was set as the Δ Ct that gave high (>90%) specificity with samples from patients not expected to have bladder cancer (data not shown). A Δ Ct above the Δ Ct cutoff for any one of the markers was considered a positive result, indicative of the presence of bladder cancer. For CRH, Ct values were used instead of Δ Ct to determine positivity for the CRH marker. A CRH Ct value <45 was considered a positive result, indicative of the presence of bladder cancer. In addition to the four bladder cancer markers (KRT20, IGF2, CRH, and ANXA10), the GeneXpert® bladder cancer assay included two controls: primers and probe for detecting ABL mRNA in the samples, and primers and probe for detecting an Armored RNA® exogenous control RNA.

(182) In the first analysis, 132 samples collected from patients who had positive cystoscopy results for bladder cancer were tested with the GeneXpert® bladder cancer assay. Sixty of those samples had also been tested using UroVysion®. Table 7 shows the results for those 132 samples.

(183) TABLE-US-00008 TABLE 7 Assay sensitivity by bladder cancer stage and grade Xpert Bladder Invalid/ UroVysion POS NEG Error** Sensitivity POS NEG inconclusive Sensitivity Stage: All 94 35 3 72.9% 29 26 5 52.7% Ta, Grade Low 29 23 2 55.8% 5 18 4 21.7% Ta, Grade High 20 2 90.9% 4 6 1 40.0% T1 13 2 86.7% 6 1 85.7% T2 11 2 84.6% 4 0 100.0% T3 3 0 100.0% 1 0 100.0% T4 2 0 100.0% CIS 11 1 91.7% 7 0 100.0% UNK 5 5 1 50.0% 2 1 66.7% Grade: All 94 35 3 72.9% 29 26 5 52.7% Low Grade 33 28 2 54.1% 6 19 4 24.0% High Grade 61 7 1 89.7% 23 7 1 76.7% **Two invalid results were due to low ABL Ct, suggesting a low number of cells in the sample, and one of the invalid results was due to poor sample quality.

(184) As shown in Table 7, for these samples, the GeneXpert® bladder cancer assay had a sensitivity of 54.1% for low grade bladder cancer and a sensitivity of 89.7% for high grade bladder cancer. In contrast, UroVysion® had a sensitivity of just 24% for low grade bladder cancer and a sensitivity of 76.6% for high grade bladder cancer. Further, the GeneXpert® bladder cancer assay was able to detect all grades and stages of bladder cancer.

(185) The same data set was then divided according to three patient groups: (A) patients with a history of bladder cancer who were currently being monitored for recurrence of bladder cancer, (B) patients who had been treated with Bacillus Calmette-Guerin (BCG) within the three months prior to sample collection, and (C) patients who were symptomatic for bladder cancer and had no prior history of bladder cancer. The results for those patient groups are shown in Table 8.

(186) TABLE-US-00009 TABLE 8 Assay sensitivity by patient population Xpert Bladder* UroVysion Grade: POS NEG Invalid/Error** Sensitivity POS NEG inconclusive Sensitivity Monitoring (Population A) All 50 20 1 71.4% 11 12 47.8% Low Grade 21 18 53.8% 2 11 1 15.4% High Grade 29 2 1 93.5% 9 1 90.0% Treated with BCG in last 3 months (Population B) All 4 2 66.7% 1 100.0% Low Grade High Grade 4 2 66.7% 1 100.0% Symptomatic (Population C) All 40 13 2 75.5% 17 14 4 54.8% Low Grade 12 10 2 54.5% 4 8 3 33.3% High Grade 28 3 90.3% 13 6 1 68.4% **Two invalid results were due to low ABL Ct, suggesting a low number of cells in the sample, and one of the invalid results was due to poor sample quality.

(187) As shown in Table 8, the GeneXpert® bladder cancer assay had a similar sensitivity for low grade and high grade bladder cancer in patients being monitored for bladder cancer and in patients who were symptomatic of bladder cancer as in the patient group as a whole (see Table 7). In patients who had been treated with BCG within the last three months, the GeneXpert® bladder cancer assay had a sensitivity of 66.7%, although the sample size was too small (6 samples) to draw any conclusions from that result.

(188) In order to have a direct comparison of the GeneXpert® bladder cancer assay and UroVysion®, a dataset was selected that included only samples that had been tested with both assays. Table 9 shows the results for that dataset, with the patients separated into two groups: (A&B) patients with a history of bladder cancer who were currently being monitored for recurrence of bladder cancer, combined with patients who had been treated with Bacillus Calmette-Guerin (BCG) within the three months prior to sample collection (these groups were combined because only one sample from a BCG-treated patient had been tested with both assays), and (C) patients who were symptomatic for bladder cancer and had no prior history of bladder cancer.

(189) TABLE-US-00010 TABLE 9 Assay sensitivity for samples tested with both GeneXpert® and UroVysion® Xpert Bladder UroVysion Grade: POS NEG Invalid/Error** Sensitivity POS NEG inconclusive Sensitivity Monitoring and BCG treated (Populations A&B) All 18 7 1 72.0% 12 13 1 48.0% Low Grade 6 7 1 46.2% 2 12 14.3% High Grade 12 0 100.0% 10 1 1 90.9% Symptomatic (Population C) All 26 8 1 76.5% 17 14 4 54.8% Low Grade 7 7 1 50.0% 4 8 3 33.3% High Grade 19 1 95.0% 13 6 1 68.4%

(190) As shown in FIG. 9, for those samples that have been tested with both GeneXpert® bladder cancer assay and UroVysion®, the GeneXpert® bladder cancer assay showed greater sensitivity than UroVysion® for detecting low grade and high grade cancer in both patient groups.

(191) Next, the data set was divided into samples that had been archived, meaning they were tested

more than one week after collection (the samples ranged from 8 days old up to nine months old), and samples that were fresh, meaning they were tested within one week of collection. The results of that analysis are shown in Table 10.

(192) TABLE-US-00011 TABLE 10 Assay sensitivity in archived and fresh samples Xpert Bladder Grade: n POS NEG Invalid/Error** Sensitivity Archived Samples All 89 61 25 3 70.9% Low Grade 46 23 21 2 52.3% High Grade 43 38 4 1 90.5% Fresh Samples All 43 33 10 76.7% Low Grade 17 10 7 58.8% High Grade 26 23 3 88.5% **Two invalid results were due to low ABL Ct, suggesting a low number of cells in the sample, and one of the invalid results was due to poor sample quality.

(193) As shown in Table 10, the GeneXpert® bladder cancer assay had a similar sensitivity for detecting low grade and high grade bladder cancer in fresh and archived samples.

7.3. Example 3: Assay Specificity for Detecting Bladder Cancer Using Marker Panel KRT20, IGF2, CRH, and ANXA10 in a Larger Cohort

(194) In addition to the samples from patients with positive cystoscopy results for bladder cancer, urine samples were collected at the seven sites from patients with negative cystoscopy results for bladder cancer, but who were being monitored for recurrence of bladder cancer, had received BCG within the three months prior to sample collection, and who appeared to be symptomatic for bladder cancer but had no prior history of bladder cancer. In addition, urine samples were collected from patients with urology referrals for other suspected conditions, such as kidney stones. Finally, urine samples were collected from healthy individuals. Urine samples were preserved and analyzed using the GeneXpert® bladder cancer assay as described in Example 2.

(195) For the samples from patients with negative cystoscopy results for bladder cancer, the assay results were divided according to the three patient groups: (A) patients with a history of bladder cancer who were currently being monitored for recurrence of bladder cancer, (B) patients who had been treated with Bacillus Calmette-Guerin (BCG) within the three months prior to sample collection, and (C) patients who were symptomatic for bladder cancer and had no prior history of bladder cancer. The results for those patient groups are shown in Table 11.

(196) TABLE-US-00012 TABLE 11 Assay specificity in cystoscopy negative patients by population Xpert Bladder POS NEG Invalid/Error Specificity Monitoring (Population A) 55 156 17 73.9% BCG Treated (Population B) 1 6 0 85.7% Symptomatic (Population C) 16 86 10 84.3%

(197) As shown in Table 11, the GeneXpert® bladder cancer assay had a specificity of 73.9%, 85.7%, and 84.3% for patient groups (A), (B), and (C), respectively.

(198) Next, the specificity of the GeneXpert® bladder cancer assay in patients who were suspected of having other urological conditions, but not bladder cancer, was determined. Seventy patient samples were collected in this category. The results with the GeneXpert® bladder cancer assay were 14 positives, 50 negatives, and 6 invalid results. The specificity of the GeneXpert® bladder cancer assay for this patient population was therefore 78.1%.

(199) Finally, the specificity of the GeneXpert® bladder cancer assay in healthy individuals was determined. Fifty-five samples were collected in this category. The results were 4 positives and 51 negatives, indicating a specificity of 92.7% for this subject category.

(200) All publications, patents, patent applications and other documents cited in this application are hereby incorporated by reference in their entireties for all purposes to the same extent as if each individual publication, patent, patent application or other document were individually indicated to be incorporated by reference for all purposes.

(201) While various specific embodiments have been illustrated and described, it will be appreciated that changes can be made without departing from the spirit and scope of the invention(s).

(202) TABLE-US-00013 TABLE OF CERTAIN SEQUENCES SEQ ID NO Description
Sequence 1 Human CRH TCGTTCCTTG GCAGGGCCCT ATGATTTATG
CAGGAGCAGA GGCAGCACGC mRNA AATCGAGCTG TCAAGAGAGC

GTACAGCTTAT	TAGGCAAATG	CTCGTGTTT	TTTGAAGAG	GTCGACATA
TAAAATCCCA	CTCCAGGCTC	TGGAGTGGAG	AAACTCAGAG	ACCAAGTCCA
TTGAGAGACT	GAGGGGAAAG	AGAGGAGAGA	AAGAAAAAGA	GAGTGGGAAC
AGTAAAGAGA	AAGGAAGACA	ACCTCCAGAG	AAAGCCCCCG	GAGACGTCTC
TCTGCAGAGA	GGCGGCAGCA	CCCGGCTCAC	CTGCGAAGCG	CCTGGGAAGC
GAGTGCCCT	AACATGCGGC	TGCCGCTGCT	TGTGTCCGCG	GGAGTCCTGC
TGGTGGCTCT	CCTGCCCTGC	CCGCCATGCA	GGGCGCTCCT	GAGCCGCGGG
CCGGTCCCGG	GAGCTCGGCA	GGCGCCGCAG	CACCCTCAGC	CCTTGGATTT
CTTCCAGCCG	CCGCCGCAGT	CCGAGCAGCC	CCAGCAGCCG	CAGGCTCGGC
CGGTCCTGCT	CCGCATGGGA	GAGGAGTACT	TCCTCCGCCT	GGGGAACCTC
AACAAGAGCC	CGGCCGCTCC	CCTTTCGCCC	GCCTCCTCGC	TCCTCGCCGG
AGGCAGCGGC	AGCCGCCCTT	CGCCGAACA	GGCGACCGCC	AACTTTTTCC
GCGTGTTGCT	GCAGCAGCTG	CTGCTGCCTC	GGCGCTCGCT	CGACAGCCCC
GCGGCTCTCG	CGGAGCGCGG	CGCTAGGAAT	GCCCTCGGCG	GCCACCAGGA
GGCACCGGAG	AGAGAAAGGC	GGTCCGAGGA	GCCTCCCATC	TCCCTGGATC
TCACCTTCCA	CCTCCTCCGG	GAAGTCTTGG	AAATGGCCAG	GGCCGAGCAG
TTAGCACAGC	AAGCTCACAG	CAACAGGAAA	CTCATGGAGA	TTATTGGGAA
ATAAAACGGT	GCGTTTGGCC	AAAAAGAATC	TGCATTTAGC	ACAAAAAAA
TTTAAAAAAA	TACAGTATTC	TGTACCATAG	CGCTGCTCTT	ATGCCATTTG
TTTATTTTTA	TATAGCTTGA	AACATAGAGG	GAGAGAGGGA	GAGAGCCTAT
ACCCCTTACT	TAGCATGCAC	AAAGTGTATT	CACGTGCAGC	AGCAACACAA
TGTTATTCGT	TTTGTCTACG	TTTAGTTTCC	GTTTCCAGGT	GTTTATAGTG
GTGTTTTTAA	GAGAATGTAG	ACCTGTGAGA	AAACGTTTTG	TTTGAAAAAG
CAGACAGAAG	TCACTCAATT	GTTTTTGTG	TGGTCTGAGC	CAAAGAGAAT
GCCATTCTCT	TGGGTGGGTA	AGACTAAATC	TGTAAGCTCT	TTGAAACAAC
TTTCTCTTGT	AAACGTTTCA	GTAATAAAAC	ATCTTTCCAG	TCCTTGGTCA
GTTTGGTTGT	GTAAGAGAAT	GTTGAATACT	TATATTTTTA	ATAAAAGTTG
CAAAGGTAAT	CATG 2 Human	IGF2 CCGCTAATGT	ACCATGCCCT	
GGTGCTGGAA	AGTGCCTGAG	CCAGCTGCCC mRNA,	transcript CAGCGGCCTC	
AGCACTACCA	AGTTGGCACA	AAGCTCCCCA	AATTCGGAGG variant 2	
GGCTCAGGGA	AACGAGTGGA	GGGGATGAGG	AGGTGAGGGG	TAAACCCATC
ATTTCAGTTG	GCATTTGAGC	AGGTGCCATG	CTCAGCGGAG	ATGAGGCTCT
CCCATCTGTA	GGGGCCGTAT	TAACATGCAC	ACTCTAAAAG	TGCCCTTCGT
TTCTCCAGCC	TCAGCTTTGT	CCCTCTCCTC	CTCCACGTCA	ACCTGGCCAG
AGGGTCTGGA	CGCCACAGCC	AGGGCACCCC	CTGCTTTGGT	GGTGACTGCT
AATATTGGCC	AGGCCGGCGG	ATCATCGTCC	AGGCAGTTTC	GGCAGAGAGC
CTTGGGCACC	AGTGACTION	CGGTCCTCTT	TATCCACTGT	CCAGGAGCTG
CGGGGACTGC	GCAGGGACTA	GAGTACAGGG	GCCGAAGAGT	CACCACCGAG
CTTGTGTGGG	AGGAGGTGGA	TTCCAGCCCC	CAGCCCCAGG	GCTCTGAATC
GCTGCCAGCT	CAGCCCCCTG	CCCAGCCTGC	CCCACAGCCT	GAGCCCCAGC
AGGCCAGAGA	GCCCAGTCCT	GAGGTGAGCT	GCTGTGGCCT	GTGGCCCAGG
CGACCCCAGC	GCTCCCAGAA	CTGAGGCTGG	CAGCCAGCCC	CAGCCTCAGC
CCCAACTGCG	AGGCAGAGAG	ACACCAATGG	GAATCCCAAT	GGGGAAGTCG
ATGCTGGTGC	TTCTCACCTT	CTTGGCCTTC	GCCTCGTGCT	GCATTGCTGC
TTACCGCCCC	AGTGAGACCC	TGTGCGGCGG	GGAGCTGGTG	GACACCCTCC
AGTTCGTCTG	TGGGGACCGC	GGCTTCTACT	TCAGCAGGCC	CGCAAGCCGT
GTGAGCCGTC	GCAGCCGTGG	CATCGTTGAG	GAGTGCTGTT	TCCGCAGCTG
TGACCTGGCC	CTCCTGGAGA	CGTACTGTGC	TACCCCCGCC	AAGTCCGAGA
GGGACGTGTC	GACCCCTCCG	ACCGTGCTTC	CGGACAACCT	CCCCAGATAC
CCCGTGGGCA	AGTTCTTCCA	ATATGACACC	TGGAAGCAGT	CCACCCAGCG

CCTGCGCAGG	GGCCTGCCTG	CCCTCGCTCG	TGCCCGCCGG	GGTCACGTGC
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CATGAAAAAA	AAAAAAAAAA	AA 4 Human	IGF2	GGCCGCGCGC
CCTCAGGACG	TGGACAGGGA	GGGCTTCCCC	GTGTCCAGGA	mRNA, transcript
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variant 3	GGGTCTTCCA	ACGGA	ACTGGG	CGTTGCTCCC
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GGATTTCAGT	CGCAGAAGCT	TCCACAGAAG	CCTGAGCTCC	TCCTTGCAGG
CCCCTGTAGT	CAGTACAGTG	GGCATGCAGC	GCCTCGGGAC	GACACCCAGC
GTTTATGGGG	GTGCTGGAGG	CCGGGGCATC	CGCATCTCCA	ACTCCAGACA
CACGGTGAAC	TATGGGAGCG	ATCTCACAGG	CGGCGGGGAC	CTGTTTGTTG
GCAATGAGAA	AATGGCCATG	CAGAACCTAA	ATGACCGTCT	AGCGAGCTAC
CTAGAAAAGG	TGCGGACCCT	GGAGCAGTCC	AACTCCAAAC	TTGAAGTGCA
AATCAAGCAG	TGGTACGAAA	CCAACGCCCC	GAGGGCTGGT	CGCGACTACA
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CAACTGCAAA	ATGCTCGGTG	TGTCCTGCAA	ATTGATAATG	CTAAACTGGC
TGCTGAGGAC	TTCAGACTGA	AGTATGAGAC	TGAGAGAGGA	ATACGTCTAA
CAGTGGAAGC	TGATCTCCAA	GGCCTGAATA	AGGTCTTTGA	TGACCTAACC
CTACATAAAA	CAGATTTGGA	GATTCAAATT	GAAGAACTGA	ATAAAGACCT
AGCTCTCCTC	AAAAAGGAGC	ATCAGGAGGA	AGTCGATGGC	CTACACAAGC
ATCTGGGCAA	CACTGTCAAT	GTGGAGGTTG	ATGCTGCTCC	AGGCCTGAAC
CTTGGCGTCA	TCATGAATGA	AATGAGGCAG	AAGTATGAAG	TCATGGCCCA

GAAGAACCTT	CAGAGGACCA	AAGAGACGTT	TGAGAGACAG	ACTGACGTTT
TGCAGCAACA	GGTCACAGTG	AATACTGAAG	AATTAAGAGG	AACTGAGGTT
CAACTAACGG	AGCTGAGACG	CACCTCCCAG	AGCCTTGAGA	TAGAACTCCA
GTCCCATCTC	AGCATGAAAG	AGTCTTTGGA	GCACACTCTA	GAGGAGACCA
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CGAATACCAT	ATCCTTCTTG	ACATAAAGAC	TCGACTTGAA	CAGGAAATTG
CTACTTACCG	CCGCCTTCTG	GAAGGAGAAG	ACGTAAAAAC	TACAGAATAT
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ACTCAGAGGA	GCTGCAAATT	GCCTGCAAAA	ATGAAATCCA	GTGAGCACTA
GAATATTTAA	AACATCATT	CTGCCATCTT	TATCATGAAG	CACATCAATT
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TGCAATACAT	TTCAATTATA	CTAAACCTCA	CAAAGTAGAG	GAATCCATGT
AAATTGCAAA	TAAACCACTT	TCTAATTTTT	TCCTGTTTCT	GAATTGTAAA
ACCCCTTTTG	GGAGTCCCTG	GTTTCTTATT	GAGCCAATTT	CTGGG 6 Human
ANXA10	ATCCAGATTT	GCTTTTACAT	TTTCTTGCCT	GAGTCTGAGG
TGAACAGTGA	mRNA ACATATTTAC	ATTTGATTTA	ACAGTGAACC	TTAATTCTTT
CTGGCTTCAC	AGTGAAACAA	GTTTATGCAA	TCGATCAAAT	ATTTTCATCC
CTGAGGTAA	CAATTACCAT	CAAAATGTTT	TGTGGAGACT	ATGTGCAAGG
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TAGGAGGAGC	ACTCCAAGGA	TTTGACTGTG	ACAAAGACAT	GCTGATCAAC
ATTCTGACTC	AGCGCTGCAA	TGCACAAAGG	ATGATGATTG	CAGAGGCATA
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GAATTGCCTC	ATTGAAATAC	TAGCTTCAAG	AACAAATGGA	GAAATTTTCC
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ACAACAA 7 Human	ABL	GGTTGGTGAC	TTCCACAGGA	AAAGTTCTGG

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AACCGCGAAA	ATAATTGGAA	CTCCTGCTTG	CAAGTGTCAA
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			CCCTCCGGGA

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CGGCTGTGAC	GCAGCCGGAG	GGAGGCACTA	GTCACCGACA	GCGGCCTTGA
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TATCTCTTGG	TATGCATCTT	TTATAGACGC	TCTTTTCTAA	GTGGCGTGTG
CATAGCGTCC	TGCCCTGCCC	CCTCGGGGGC	CTGTGGTGGC	TCCCCCTCTG
CTTCTCGGGG	TCCAGTGCAT	TTTGTTCCTG	TATATGATTC	TCTGTGGTTT
TTTTTTGAATC	CAAATCTGTC	CTCTGTAGTA	TTTTTTAAAT	AAATCAGTGT
TTACATTAGA	A 8 Armored RNA	® GAUGCUCACU	UCAUCUAUGG	
UUACCCUGGG	ACUUUUACAC	CAACAGAACU	sequence AGCAUCAUCC	
UCUGCAUGGU	CAGGUCAUGG	AUCGGCAUCC	UGACAGUUUC	GGGAAUUAGG
CAUCUGCAGU	CUUACUGCUC	AUCGGCUGAU	GAUGCUGCUG	UAAUCCCCAU
CCAAGCAAGC	UUGUGAUCCU	CCGCCAUUUA	CCCAAUUGGU	AUAACAUUUA
GGACUUAAAG	CUAUGCAAUU	AUCACCUUGU	UUUUAACAG	CAAGACCUAA
UAUUUUCUUU	UCAUCAUUAA	UGCCUUUUGA	UGGAUCAGGC	AACCAUUUAU
AAAU AUGUUC	ACCAGCCGAA	GUCAGUAGUG	AUUGGGUGGU	UCCUGGCUUG
GGAUCAUGCC	GCUGCAGAGG	CUAUUCUCCU	CUUGGCAGAU	UGUCUGUAGC
CGAGAAGGCG	GAGUCUGGCA	AUGAUCAUGC	AUACAGUGUA	CGACAGCCUU
AGGGACUGGA	GCUCAAGCAG	UGUUUCCUCA	ACCAGUCACA	

Claims

1. A method for detecting bladder cancer markers in a subject comprising detecting the levels of each bladder cancer marker amplicon of a set of bladder cancer markers in a urine sample or bladder washing sample from the subject, wherein the set of bladder cancer markers consists of corticotrophin releasing hormone (CRH), insulin-like growth factor 2 (IGF2), keratin 20 (KRT20) and annexin 10 (ANXA10), by contacting RNA from the sample with a set of primers, conducting one or more polymerase chain reactions (PCR) and detecting a set of bladder cancer marker amplicons that is produced by the PCR.
2. The method of claim 1, wherein the method further comprises detecting an endogenous control and/or exogenous control.
3. The method of claim 2, wherein the endogenous control is selected from ABL, GUSB, GAPDH, TUBB, and UPK1.
4. The method of claim 2, wherein the exogenous control is an RNA.
5. The method of claim 2, wherein the method comprises comparing a Ct value or a Δ Ct value to a threshold Ct value or Δ Ct value, wherein Δ Ct is the Ct value for the control minus the Ct value for the marker.
6. The method of claim 1, wherein the detecting comprises RT-PCR.
7. The method of claim 1, wherein the detecting comprises a RT-PCR reaction that takes less than 2 hours from an initial denaturation step through a final extension step.
8. The method of claim 1, wherein the set of primers comprises a first and second primer for

detecting CRH comprising: a) a first primer comprising at least 12 contiguous nucleotides of SEQ ID NO: 19 and a second primer comprising at least 12 contiguous nucleotides of SEQ ID NO: 20, wherein each primer is less than 50, less than 45, less than 40, less than 35, or less than 30 nucleotides long; or b) a first primer comprising at least 12 contiguous nucleotides of SEQ ID NO: 35 and a second primer comprising at least 12 contiguous nucleotides of SEQ ID NO: 36, wherein each primer is less than 50, less than 45, less than 40, less than 35, or less than 30 nucleotides long.

9. The method of claim 1, wherein the set of primers comprises a first and second primer for detecting IGF2 comprising: a) a first primer comprising at least 12 contiguous nucleotides of SEQ ID NO: 16 and a second primer comprising at least 12 contiguous nucleotides of SEQ ID NO: 17, wherein each primer is less than 50, less than 45, less than 40, less than 35, or less than 30 nucleotides long; or b) a first primer comprising at least 12 contiguous nucleotides of at least 12 contiguous nucleotides of SEQ ID NO: 32 and a second primer comprising SEQ ID NO: 33, wherein each primer is less than 50, less than 45, less than 40, less than 35, or less than 30 nucleotides long.

10. The method of claim 1, wherein the set of primers comprises a first and second primer for detecting KRT20 comprising: a) a first primer comprising at least 12 contiguous nucleotides of SEQ ID NO: 13 and a second primer comprising at least 12 contiguous nucleotides of SEQ ID NO: 14, wherein each primer is less than 50, less than 45, less than 40, less than 35, or less than 30 nucleotides long; or b) a first primer comprising at least 12 contiguous nucleotides of SEQ ID NO: 29 and a second primer comprising at least 12 contiguous nucleotides of SEQ ID NO: 30, wherein each primer is less than 50, less than 45, less than 40, less than 35, or less than 30 nucleotides long.

11. The method of claim 1, wherein the set of primers comprises a first and second primer for detecting ANXA10 comprising: a) a first primer comprising at least 12 contiguous nucleotides of SEQ ID NO: 26 and a second primer comprising at least 12 contiguous nucleotides of SEQ ID NO: 27, wherein each primer is less than 50, less than 45, less than 40, less than 35, or less than 30 nucleotides long; or b) a first primer comprising at least 12 contiguous nucleotides of SEQ ID NO: 38 and a second primer comprising at least 12 contiguous nucleotides of SEQ ID NO: 39, wherein each primer is less than 50, less than 45, less than 40, less than 35, or less than 30 nucleotides long; or c) a first primer comprising at least 12 contiguous nucleotides of SEQ ID NO: 48 and a second primer comprising at least 12 contiguous nucleotides of SEQ ID NO: 39, wherein each primer is less than 50, less than 45, less than 40, less than 35, or less than 30 nucleotides long.

12. The method of claim 1, wherein the set of bladder cancer marker amplicons comprises a CRH amplicon, an IGF2 amplicon, a KRT20 amplicon, and an ANXA10 amplicon, and wherein the method comprises contacting the bladder cancer marker amplicons with a set of bladder cancer marker probes, wherein the set of bladder cancer marker probes comprises a probe for detecting the CRH amplicon, a probe for detecting the IGF2 amplicon, a probe for detecting the KRT20 amplicon, and a probe for detecting the ANXA10 amplicon.

13. The method of claim 12, wherein the probe for detecting CRH comprises at least 12 contiguous nucleotides of SEQ ID NO: 21, SEQ ID NO: 22, or SEQ ID NO: 37, wherein the probe is less than 50, less than 45, less than 40, less than 35, or less than 30 nucleotides long.

14. The method of claim 12, wherein the probe for detecting IGF2 comprises at least 12 contiguous nucleotides of SEQ ID NO: 34 or at least 12 SEQ ID NO: 18, wherein the probe is less than 50, less than 45, less than 40, less than 35, or less than 30 nucleotides long.

15. The method of claim 12, wherein the probe for detecting KRT20 comprises at least 12 contiguous nucleotides of SEQ ID NO: 15 or SEQ ID NO: 31, wherein the probe is less than 50, less than 45, less than 40, less than 35, or less than 30 nucleotides long.

16. The method of claim 12, wherein the fourth probe for detecting ANXA10 comprises at least 12 contiguous nucleotides of SEQ ID NO: 28 or SEQ ID NO: 40, wherein the probe is less than 50, less than 45, less than 40, less than 35, or less than 30 nucleotides long.

17. The method of claim 12, wherein each bladder cancer marker probe comprises a dye, and

wherein each dye is detectably different from the other dyes.

18. The method of claim 17, wherein each probe comprises a quencher molecule.

19. The method of claim 12, wherein the method further comprises forming an endogenous control amplicon, and contacting the endogenous control amplicon with an endogenous control probe, and/or forming an exogenous control amplicon, and contacting the exogenous control amplicon with an exogenous control probe, wherein each probe comprises a dye, and wherein each dye is detectably different from the other dyes.

20. The method of claim 1, wherein the set of bladder cancer markers are detected in a single multiplex reaction.

21. The method of claim 1, wherein the subject has at least one symptom of bladder cancer and/or has a history of bladder cancer.
